

Dendritic morphology as a dictionary for local credit assignment

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Abstract

Dendritic trees compartmentalize synaptic inputs, but which spatial distinctions a learning signal must preserve is unclear. We describe delivered credit as combinations of spatial profiles constrained by morphology. In image classifiers with one dendritic neuronal layer, preserving neuronal identity accounts for most of the feedback benefit. Contextual conflict favors branch selection, while hierarchical distractors favor ancestry grouping at a bandwidth predicted by the task. Targets with matched input-sensitivity spectra develop different learned credit spectra and tolerance of fixed profiles; a local inhibitory gate also supports learning of opposing branch tuning in a conductance tree. Serial compartments help when nuisance-factor information is distributed across modules. Reconstructed arbors compress modeled passive perturbation fields through ancestry profiles, and focal shunts regulate their gains with selectivity set by electrotonic state. Measured visual responses provide no evidence of preferential ancestry alignment in the sampled cells. The framework links task structure to useful credit distinctions while separating available delivery patterns from the mechanisms that generate their coefficients.

Introduction

Learning must link a synapse’s local activity to its contribution to a circuit outcome. Cable filtering, nonlinear integration and spatially organized inhibition make locations within a dendritic tree functionally distinct [1–7]. These differences expand computation, but when does branching also help distribute useful learning signals?

Credit assignment asks how changing a synapse would affect task error. Backpropagation (BP) answers this by calculating the loss gradient—the sensitivity of task error to each parameter [8]; related sensitivity calculations apply to recurrent and steady-state networks [9, 10]. For a point neuron, each incoming-weight gradient is the product of a locally available sensitivity, called eligibility, and a neuron-specific learning signal. Hebbian plasticity captures the association between local pre- and postsynaptic variables [11], while three-factor rules combine that eligibility with an outcome-dependent signal [12]. Here, a rule is local when a synapse uses its own eligibility and an available learning signal. This definition does not require that the signal itself be computed locally.

Biologically motivated learning theories differ in how they obtain such signals. Feedback alignment communicates errors through separate pathways [13, 14]; equilibrium, predictive and prospective methods infer target states or errors from dynamics [15–17]; eligibility propagation combines temporal traces with neuron-specific signals [18]. One number shared across neurons carries less task information than separately assigned neuronal signals, although each synapse still has its own eligibility [19–21]. Recent causal evidence for distinct instructive signals across neuronal populations motivates asking what additional distinctions are useful within a neuron’s arbor [22].

We describe this organization using a *route dictionary*: a set of spatial profiles for delivering learning signals. A route identifies the locations reached by a signal; its profile specifies their relative amplitudes, and a coefficient scales that profile on a given trial. Each dictionary column stores one profile. A shared profile broadcasts throughout an arbor, while several profiles can distinguish subtree addresses. Their linear combinations give the available fields. We use *basis* for linearly independent profiles and *dictionary* when profiles may be redundant. Morphology constrains the available routes; the task and teaching mechanism determine useful coefficients.

Conductance changes the gains of anatomical delivery routes. Excitatory and inhibitory contacts open nonnegative conductances whose reversal potentials determine current; inhibition can lower input resistance and attenuate other

inputs even when its own current is small [23–27]. Rescaling a delivery channel preserves its spatial profile, whereas altered cable transport can reshape how local voltage affects somatic output and task error.

Dendritic models already connect nonlinear compartments and structural plasticity to memory capacity [28], and branch-specific plasticity can organize nonlinear feature combinations [29]. Successive computations in artificial trees and detailed cortical models also establish substantial forward capacity [30, 31]. Dendritic learning theories supply mechanisms for somatic prediction, segregated teaching, interneuron-mediated errors, bursts and prospective dynamics [32–38]; cable-learning models optimize reconstructed neurons using synaptic sensitivities [39]. Our question concerns the spatial information these learning mechanisms must preserve across an arbor.

The conductance-tree gradient factorization and LocalCA rule were introduced in our predecessor preprint [40]. This Article extends those foundations with controlled task-to-credit comparisons, matched-spectrum and conductance learning experiments, and anatomical transport and response tests.

We compare the value of preserving neuronal identity, branch selection, ancestry grouping and example-dependent credit across interacting branches. Morphology supplies candidate patterns and conductance regulates their expression. We establish forward compatibility before interpreting learning gaps; measured responses test the anatomical proposal. The question is which distinctions useful credit must preserve, rather than whether finer feedback is always better.

Results

Exact gradients separate local sensitivity from downstream credit

A useful synaptic update combines local sensitivity with the effect that the change would have on task error. For a point neuron $y_u = f_u(\sum_i w_i x_i + b_u)$, the chain rule gives $\partial \mathcal{L} / \partial w_i = x_i f'_u \delta_u$, where x_i and w_i are synaptic input and weight, b_u is bias, and $\delta_u = \partial \mathcal{L} / \partial y_u$ is the neuronal error. The derivative f'_u is evaluated at the current input sum. Thus local eligibility $x_i f'_u$ and task-derived credit supply different information (Fig. 1A).

We replace the point neuron by a rooted tree with spatially distinct addresses, with soma $n = 0$, compartment voltage V_n , and nonnegative synaptic conductances g_i . A contact with reversal potential E_i^{rev} and nonnegative activity x_i contributes current $g_i x_i (E_i^{\text{rev}} - V_n)$. In the directed artificial trees, child outputs drive their parents and a terminal-to-soma pass computes the steady state. Total input resistance R_n^{tot} is the reciprocal of the summed leak, excitatory, inhibitory and child-coupling conductances (Methods). Differentiating the current balance yields

$$\frac{\partial \mathcal{L}}{\partial g_i} = \underbrace{x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n)}_{\text{local eligibility}} \underbrace{\delta_{n,u}^V}_{\partial \mathcal{L} / \partial V_n}. \quad (1)$$

Both excitatory (E) and inhibitory (I) contacts obey this factorization (Fig. 1A). Eligibility tells how conductance changes local voltage; compartment credit tells how that voltage changes task loss.

Here voltage error is a loss derivative, not a voltage difference. In a directed tree, each compartment has one path to the soma. Its exact voltage error is

$$\delta_{n,u}^V = \delta_{0,u}^V \tilde{\alpha}_n, \quad \tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}. \quad (2)$$

Here $f'_i(V_i)$ is the child compartment’s output derivative, $g_{i \rightarrow k}^{\text{den}}$ is its coupling conductance to parent k , and $\delta_{0,u}^V = f'_0(V_0) \delta_u$ is the somatic voltage error. Thus the path multiplier $\tilde{\alpha}_n$ converts one somatic error into a spatial field. The conductance-only component of this product is attenuated by additional conductance on a path’s ancestors; branch nonlinearities and forward voltages can change as well (Supplementary Section S1).

Reciprocal cables instead require a joint adjoint calculation of how local currents affect task loss (Methods). These mathematical sensitivities do not establish a biological error-transmission mechanism.

Our local credit assignment (LocalCA) rule retains the local sensitivity but supplies an approximate learning signal to each compartment. A hat denotes this delivered field:

$$\widehat{V}_{g_i} \mathcal{L} = x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n) \widehat{\delta}_{n,u}^V, \quad (3)$$

$$\widehat{\delta}_u^V = A_u \mathbf{c}_u. \quad (4)$$

Columns of $A_u \in \mathbb{R}^{N_u \times K}$ are fixed spatial delivery profiles over N_u routed nonsomatic compartments; $\mathbf{c}_u$ contains their example-dependent coefficients. To distinguish coefficient generation from delivery gain, we can write $\mathbf{c}_u = \Gamma_u \mathbf{c}_u^{\text{source}}$, where diagonal Γ_u scales each profile and $\mathbf{c}_u^{\text{source}}$ is the supplied signal (Fig. 1A). Broadcast, subtree and compartment profiles give increasingly detailed spatial addresses (Fig. 1B). This linear delivery model does not follow automatically from the forward cable equations and does not specify how a circuit estimates the coefficients.

At $K = 1$, the all-ones column $A_u = \mathbf{1}$ broadcasts one coefficient through the arbor; at $K = N_u$, the identity matrix $A_u = I$ can reproduce the full nonsomatic field. Nested subtrees provide intermediate dictionaries. Profile count, rank (the number of independent profiles) and coefficient information are distinct resources. Exact path transport combines one somatic scalar with state-dependent local derivatives; it does not require an independent external error for every compartment.

Capture is the fraction of squared learning-signal magnitude that a dictionary can reproduce. For a nonzero field $\boldsymbol{\delta}$,

$$\mathcal{C}(A; \boldsymbol{\delta}) = \frac{\|P_A \boldsymbol{\delta}\|^2}{\|\boldsymbol{\delta}\|^2}, \quad P_A = AA^\dagger.$$

Here P_A gives the best least-squares reconstruction from the profiles; $A^\dagger$ is the Moore–Penrose pseudoinverse and $\|\cdot\|$ the Euclidean norm. Thus capture is a fraction of squared field magnitude, not a fraction of correct decisions. The mean of these per-field ratios weights fields equally. In contrast, $\text{tr}(P_A C) / \text{tr}(C)$, with $C = \mathbb{E}[\boldsymbol{\delta} \boldsymbol{\delta}^T]$, is the fraction of total ensemble energy captured, weighting fields by their energy. Here C is the uncentered second-moment matrix and tr denotes its trace, the sum of diagonal entries. The two capture measures need not agree. Projection chooses the best coefficients in the specified metric; an implemented learning rule need not supply them.

The expansion applies in a specified coordinate. In the nonlinear image models, we route errors of compartment outputs and retain the local output derivative in eligibility; voltage-error capture is reported separately. Identity-output conductance trees use voltage errors directly. Supplementary Table S2 compares the coordinates, coefficient sources and state dependence. Dictionary capacity answers what fields can be delivered; learning experiments determine which distinctions improve performance.

A restricted dictionary can also filter noise. In a controlled update-space model, projection removes update variation outside the chosen span. A narrower span that retains most of the useful gradient can give a better one-step guarantee when it admits less noise, despite discarding some signal (Fig. 1C). If the span contains the entire gradient, its mean is preserved. A one-step smoothness bound formalizes this balance between retained descent and update variance (Methods; Supplementary Fig. S2). Its initialization utility did not select the best final morphology: in a separate 12,800-fit prospective experiment, it lost to a fixed maximum-budget tree in all twenty seed averages in both selection arms (Supplementary Fig. S33). The local filtering principle motivates testing useful spatial distinctions without assuming that one initial score predicts the outcome of an entire learning trajectory.

One signal per neuron captures most of the image-task learning benefit

We first tested credit resolution on image classification. All image benchmark networks used one dendritic layer and a trainable linear readout. In the primary comparisons, the readout supplied exact neuronal output errors before the feedback intervention. The MNIST comparison used 128 directed $[3, 3]$ trees, each with three proximal and nine terminal compartments, receiving flattened images. Shunting and raw-additive architectures shared the graph, initial contact construction, nonlinearities and training protocol, with architecture-specific initialization. The additive control removed reversal-potential driving forces and the conductance denominator, retaining signed current summation without shunting.

114 Six rules separated neuronal identity, spatial resolution and learning in the dendritic core (Fig. 1B,D). A strict
 115 scalar shared one signal across neurons; neuron-specific feedback broadcast each neuron’s own output error through
 116 its arbor. A projected one-profile rule averaged the exact activation errors over all twelve nonsomatic compartments,
 117 while three subtree profiles averaged them within each proximal compartment and its three children. Both projections
 118 acquired coefficients from the same exact field and preserved its common-mode mean at identical parameters,
 119 although their field norms could differ. Exact transport retained every compartment error. A decoder-only reference
 120 trained the readout while freezing the core. Three development seeds selected each rule’s rate from three common
 121 multipliers; ten fresh paired seeds per architecture evaluated selected and original common rates over 180 epochs
 122 (Methods; Supplementary Fig. S7A,B).

123 At selected rates, neuron-specific feedback exceeded the strict scalar by 9.23 percentage points in shunting trees
 124 and 7.46 points in additive trees, reaching 97.12% and 97.11% accuracy. Decoder-only learning reached 80.96% and
 125 89.57%, respectively (Fig. 1D). Increasing the projected dictionary from one to three profiles changed accuracy by
 126 only 0.049 points in shunting trees (95% paired interval, -0.048 to 0.141) and 0.036 points in additive trees (-0.056
 127 to 0.122 ; Fig. 1E). The common-rate comparison gave similarly small differences (Supplementary Fig. S7C). Exact
 128 transport added 0.010 points over neuron-specific feedback in shunting trees, but 0.182 points in additive trees
 129 (0.114 – 0.252 ; positive in all ten seeds). Both exact transport and one projected profile reached about 97.3% in
 130 additive trees. Most learning benefit therefore came from preserving neuronal coordinates; the smaller additive
 131 improvement was attainable without separating subtrees.

132 A separate fifteen-seed cohort and exact-gradient checks showed spatial structure in trained fields (Supplementary
 133 Section S2 and Fig. S7E–G): moving from one to three profiles raised mean voltage-credit capture from 0.480 to
 134 0.625 in shunting trees and from 0.519 to 0.718 in additive trees. The new checkpoint analysis likewise found
 135 improved capture without a corresponding accuracy gain from subtree distinctions (Supplementary Fig. S7D). It
 136 reports activation-space capture, matching the trained projections, with voltage-space results supplied separately.
 137 These reconstructions measure available capacity, not the quality of a biological coefficient encoder. Separate
 138 recipient-permutation controls also reduced accuracy while preserving signal values and bandwidth, implicating
 139 neuronal identity (Supplementary Section S2 and Source Data).

140 With fixed random soma-level feedback, neuron-specific signals improved over scalar feedback by about eleven
 141 points, while exact paths added at most 0.09 points (fifteen paired seeds; Fig. 1F; Supplementary Fig. S6D,E). This
 142 control changes weight transport but retains the single dendritic layer and the same output-error dimensionality.
 143 Fashion-MNIST also favored neuronal feedback. On flattened CIFAR-10, exact paths reduced accuracy relative to
 144 neuronal feedback by 0.86 points (0.55 – 1.17), while remaining equivalent to matched backpropagation within the
 145 predefined margin (Fig. 1F; Supplementary Fig. S6A–C).

146 These benchmarks show that a spatially structured error field need not confer a learning advantage on finer
 147 delivery. Supplementary Figs. S6 and S8 retain further depth and task-generator controls. We next introduce task
 148 structure that makes particular spatial distinctions useful.

149 **Branch-selective information prevents updates from inactive distractors**

150 To isolate branch-selective information, we used one logistic output with $B \in \{2, 4, 8\}$ parallel branch weight
 151 vectors and no internal forward compartments or conductance nonlinearity (Fig. 2A). Context selected the branch
 152 determining the output, while every branch received an image and formed an ungated local eligibility.

153 Inputs were average-pooled, standardized Fashion-MNIST images from classes 0 and 6. The selected image
 154 determined the binary target. At conflict dose $\chi = 0$, all branches received that image; increasing χ independently
 155 replaced nonselected inputs with opposite-class images. For example i , let b_i^* be the selected branch, $\mathbf{x}_{i,b}$ its input
 156 vector and $y_i \in \{0, 1\}$ the target. The branch weights $\mathbf{w}_b$ give the pre-probability score (logit) and output error

$$z_i = \mathbf{w}_{b_i^*}^T \mathbf{x}_{i,b_i^*}, \quad \delta_i = \sigma(z_i) - y_i. \quad (5)$$

157 Here $\sigma(z) = 1/(1 + e^{-z})$ is the logistic function. The exact derivative of every unselected branch is zero, irrespective
 158 of its input. Increasing conflict therefore does not create opposing nonzero exact branch gradients. It makes a rule
 159 that ignores the selector increasingly sensitive to irrelevant, opposite-class eligibilities.

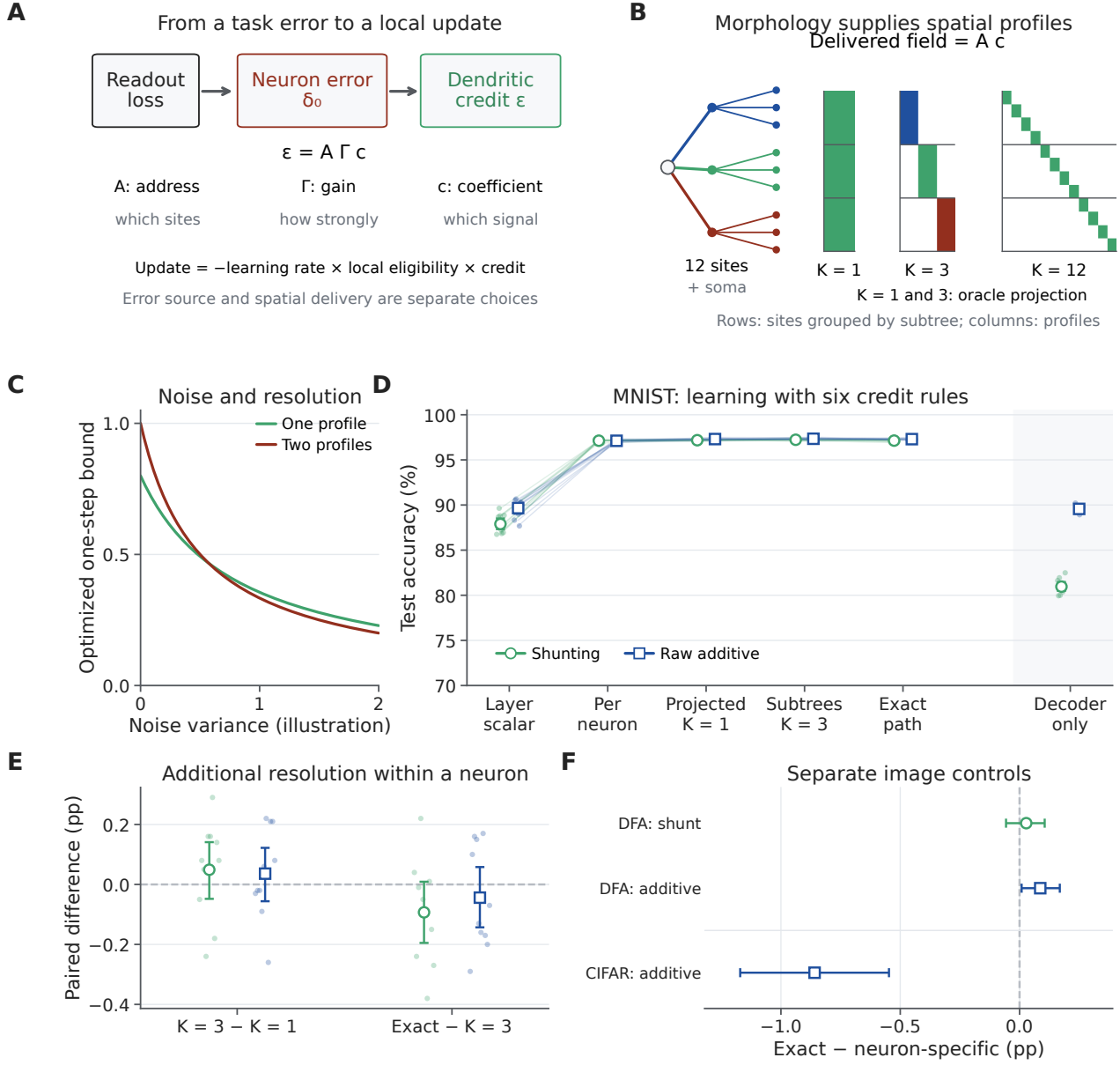


Figure 1: Task-derived credit separates neuronal identity, spatial address and gain. **A**, A neuron-specific error supplies coefficients c ; dictionary A and diagonal gains Γ give the delivered field $\epsilon = A\Gamma c$. Synaptic updates multiply credit by local eligibility and negative learning rate. **B**, Twelve nonsomatic sites with $K = 1$ common, $K = 3$ subtree or $K = 12$ site profiles. Projected $K = 1, 3$ rules use oracle activation-error coefficients and retain exact somatic errors. **C**, Illustrative projection of a unit-energy gradient with isotropic noise variance σ^2 . A rank- K span retains gradient energy q . Twice the smoothness constant L times the optimized one-step progress bound is $q^2/(q + K\sigma^2)$, comparing $(K, q) = (1, 0.8)$ and $(2, 1)$. This local bound does not predict final trained structure. **D**, MNIST with one 128-neuron dendritic layer and a linear readout. Per-neuron credit broadcasts exact somatic activation errors; decoder-only freezes the dendritic core. Lines pair ten fresh seeds per architecture. Rates were selected on three development seeds; checkpoints were validation-selected within 180 epochs. **E**, Paired effects of additional spatial resolution in the same cohort. **F**, Separate MNIST direct-feedback-alignment (DFA; fifteen seeds) and flattened CIFAR-10 (twenty seeds) controls. Symbols and intervals in **D–F** show means and descriptive 95% intervals; colors and shapes identify architectures. pp, percentage points.

For N examples, correctly selected and neuron-shared gradient estimates were

$$\mathbf{d}_b^{\text{selected}} = N^{-1} \sum_i \delta_i \mathbf{1}[b_i^* = b] \mathbf{x}_{i,b}, \quad \mathbf{d}_b^{\text{shared}} = N^{-1} \sum_i \delta_i B^{-1} \mathbf{x}_{i,b}. \quad (6)$$

The indicator $\mathbf{1}[\cdot]$ is one when its condition holds, zero otherwise; $1/B$ matches scale without supplying branch information. A feedback selector and a branch-local eligibility gate give the same update (Fig. 2B), testing access to selection rather than a need for independently computed external errors.

A simplified calculation asks when the selected branch’s evidence is outweighed by conflicting evidence sent to other branches. With centered, symmetric class evidence, the shared-mode prediction crosses zero at $\chi_c = B/[2(B-1)]$: 1, 2/3 and 4/7 for two, four and eight branches (Fig. 2C; Methods). More branches therefore predict an earlier loss of compatibility. This average calculation is distinct from the conflict dose at which a trained model reaches chance accuracy.

At zero conflict, selected and shared rules achieved similar accuracy. Their difference increased with conflict, reaching 35–58 percentage points at full conflict across the tested branch counts (20 paired seeds; Fig. 2D). In every seed, the advantage of correct routing increased with conflict. A cyclic reassignment sent every signal to the wrong branch at matched rank and sparsity. It was harmful even at zero conflict, reducing accuracy by 12–25 points relative to correct selection. Analytic backpropagation and an explicitly gated grouped-point model matched correct selection exactly, as required by their algebraic equivalence (Fig. 2E; Supplementary Section S3).

The shared rule reached chance at lower conflict as branch count increased, consistent with the predicted shift (Fig. 2F); five two-branch seeds never crossed in the sampled range. Theoretical, mean-curve and individual-seed crossings are distinct. Gradient alignment also changed during learning (Methods; Supplementary Fig. S9C–E). Credit reaching inactive branches can therefore harm learning; dendrites offer one possible selection site. A separate two-stream task tests learning and interference effects (Supplementary Fig. S9A,B).

Ancestry provides a modest advantage at an intermediate feedback budget

Branch selection compares fully shared with fully resolved feedback. We next asked whether nested subtree routes offer a useful intermediate dictionary. The task contained eight independent, eight-dimensional terminal weight blocks at the leaves of a balanced binary feedback tree. Only the context-selected block determined the logit. Internal tree nodes defined feedback supports and did not perform serial forward computation (Fig. 3A).

The selected stream carried the class signal. Distractors carried opposing signals that strengthened with distance through the feedback hierarchy, deliberately favoring task-aligned grouping. At each budget $K \in \{1, 2, 4, 8\}$, the routing field had K independent patterns and unit-length rows. Ancestry routes grouped the terminals into the whole tree, halves, sibling pairs or individual terminals. Permuting task streams across leaves changed alignment without changing the streams or forward parameter budget. Here bandwidth counts independent feedback patterns, not physical stages or independent scalar losses; other single-pattern controls need not be uniform. The complete generator and update are given in Methods.

The generator predicts where an intermediate ancestry benefit can occur. The selected stream has class coefficient one; its sibling, same-half nonsiblings and other-half streams have coefficients -0.15 , -0.45 and -0.75 . The unnormalized sum of the class signal within the selected ancestry group gives -3.05 , -0.05 , 0.85 and 1 at $K = 1, 2, 4, 8$, respectively. Thus coarse pooling reverses or nearly cancels the signal, sibling-pair routing first preserves a positive intermediate-group signal, and full resolution removes the grouping distinction. The implemented unit-row normalization divides these sums by the square root of group size, preserving their signs. This is an analytic consequence of the fixed task design; its location is not an independently discovered optimum (Fig. 3B).

Controls preserved rank while changing recipient assignment, support geometry or sparsity: cyclic derangement, depth-interleaving, random-sparse and dense random fields. The per-seed best-control summary takes their maximum after training; it is not a learned selector. A training-informed principal-component reference is retained separately (Methods).

Increasing bandwidth improved correct routing, while derangement caused large losses (Fig. 3C). Degree- and depth-preserving rewiring removed the task-aligned advantage at intermediate budgets but did not affect fully shared or fully resolved ancestry routes (Fig. 3D).

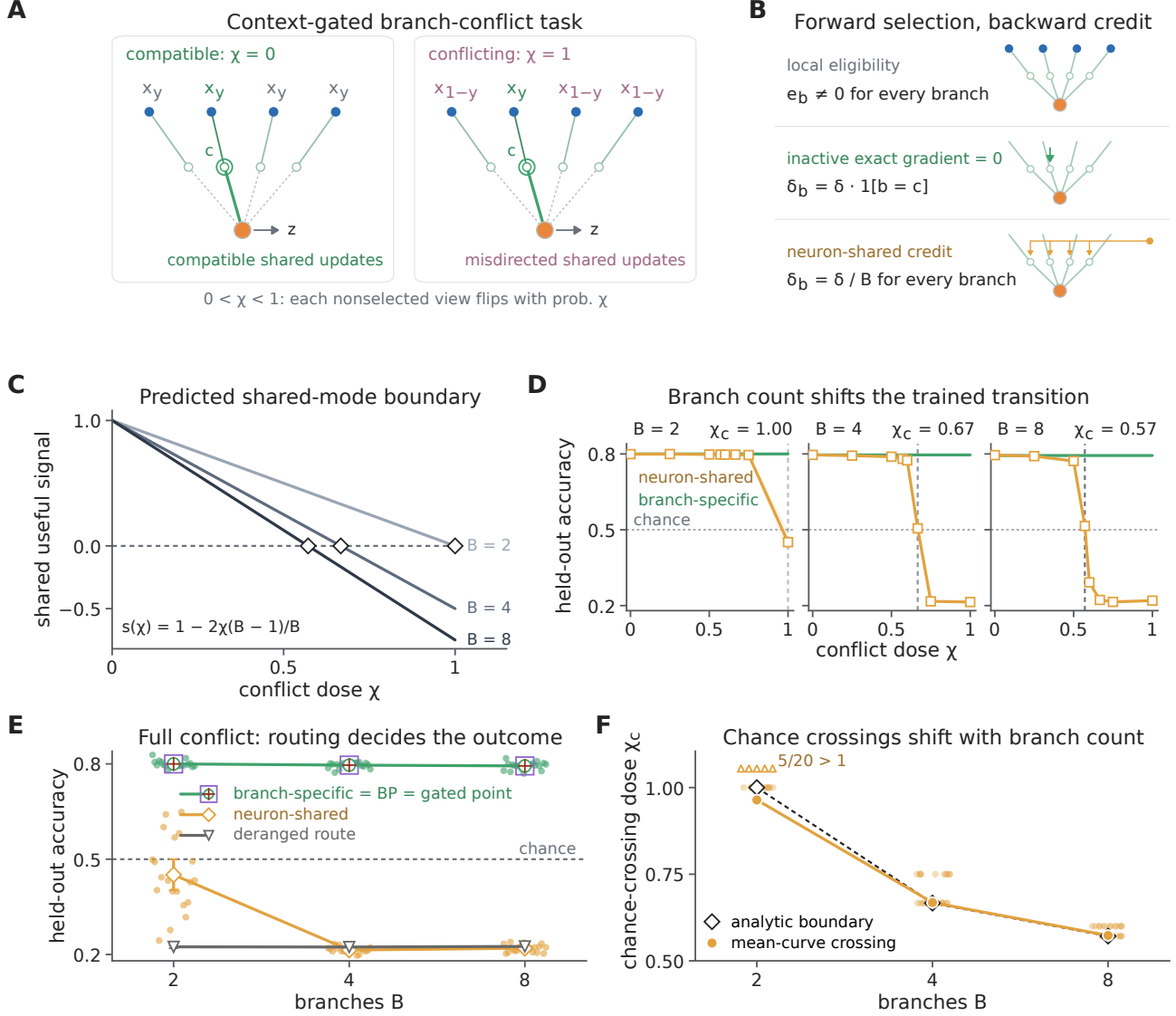


Figure 2: Branch selection prevents learning from inactive distractors. One logistic output combines B parallel branch weight vectors; there are no serial or conductance nonlinearities. **A**, Context-gated Fashion-MNIST task. Context c selects the branch determining logit z ; every branch receives an image and forms an ungated eligibility. Conflict dose χ independently replaces each nonselected image by an opposite-class image. Solid and dashed proximal paths are selected and gated-off forward routes; the double ring marks context. **B**, Selected feedback: $\delta_i \mathbf{1}[b = b_i^*]$, shared feedback: δ_i / B . Here δ_i is the output error, b_i^* the selected branch and $\mathbf{1}$ the indicator; unselected exact derivatives are zero. A local eligibility gate gives the same update. **C**, Centered symmetric-evidence prediction $s(\chi) = 1 - 2\chi(B - 1)/B$: shared-mode compatibility vanishes at $\chi_c = B/[2(B - 1)]$. This mean-field boundary is distinct from a trained chance crossing. **D**, Held-out accuracy over conflict dose. **E**, At full conflict, correct selection, backpropagation (BP) and the algebraically equivalent gated-point calculation coincide; cyclic derangement sends signals to the wrong branch at matched rank and sparsity. Selection-by-conflict interactions are positive in all 20 seeds for each B . **F**, Analytic threshold (diamonds), first interpolated crossing of mean accuracy (orange), and each seed's first sampled at-or-below-chance dose (faint dots). Open overflow triangles retain five $B = 2$ seeds that do not cross within the sweep. Curves and bars summarize 20 paired seeds with 95% seed-bootstrap intervals. Branch-count comparisons do not hold capacity fixed.

206 The ancestry-specific effect was much smaller. Ancestry performed worse than the best matched control at
 207 low budgets, exceeded it by 1.27 percentage points at $K = 4$, and tied at full resolution. The paired $K = 4$ interval
 208 was 0.59–1.95 points, with 15/20 positive differences; a paired Wilcoxon signed-rank test gave Holm-adjusted
 209 $P = 0.0101$ across four budgets. Individual control contrasts and paired seed differences distinguish this effect from
 210 the much larger correct-versus-deranged contrast (Fig. 3E,F).

211 Flat, grouped-point and gated-point representations receiving the same field behaved identically by construction.
 212 The experiment therefore supports a conditional advantage of ancestry-shaped feedback under limited bandwidth,
 213 without attributing the effect to dendritic material or treating equivalent representations as independent confirmations.

214 Learning the route coefficients was another limitation. In twenty fresh seeds, an estimator trained on separate
 215 noisy context cues improved accuracy over a frozen profile by 7.16 points, but remained 54.47 points below oracle
 216 context coefficients. Exploratory selection of its most probable route narrowed the gap, whereas delaying cues
 217 removed the benefit (Supplementary Fig. S10B–D and Section S3; Methods). These tests expose requirements for
 218 selective, timely delivery; they do not derive the context signal from local eligibility.

219 **Cross-branch interactions change the spatial credit that learning uses**

220 Interactions across branches can change credit’s sign and magnitude between examples. We compared targets with
 221 identical input-sensitivity spectra on the same representationally compatible tree, separating credit delivery from
 222 forward capacity.

223 Eight independent inputs x_i were equally likely to be -1 or $+1$. Each branch computed $u = a + bu_L + cu_R + du_Lu_R$
 224 with four trainable coefficients. This multi-affine unit is affine in either child while the other is fixed. Pairwise
 225 targets summed four disjoint products, $f_M = \frac{1}{2} \sum_{(i,j) \in M} s_{ij} x_i x_j$; quartic targets combined complementary quartets,
 226 $f_A = \frac{1}{2} (s_A \prod_{i \in A} x_i + s_{\bar{A}} \prod_{i \notin A} x_i)$. Here M partitions inputs into pairs, A is one quartet, $\bar{A}$ its complement, and signs s
 227 were independently sampled. Within each seed, targets shared a compatible balanced tree, input assignment, initial
 228 weights and examples (Fig. 4A). Both have input-sensitivity second moment $\mathbb{E}[\nabla_x f \nabla_x f^T] = I_8/4$, giving identical
 229 strengths of independent sensitivity directions.

230 Compatibility depends on what each branch must communicate to its parent. Only one scalar leaves a branch, so
 231 it cannot transmit several independently required combinations of its inputs. This gives an exact criterion: at each
 232 subtree boundary, the matrix of interaction coefficients between inside and outside input products must have rank at
 233 most one after removing the constant inside row. The condition is necessary and sufficient across all boundaries;
 234 discarded singular values bound representation error (Methods; Supplementary Section S4).

235 Logical rules make this forward constraint concrete. All fifteen binary trees over four inputs can represent AND,
 236 OR and parity exactly, but mixed rules can require particular input groupings. The rule $(a \text{ AND } b) \text{ OR } (c \text{ AND } d)$
 237 needs only two serial stages, whereas $a \text{ AND } [b \text{ OR } (c \text{ AND } d)]$ needs three in this scalar model, despite containing
 238 the same two AND gates and one OR gate. These are exact representability statements, established independently
 239 of training (Supplementary Figs. S13–S14). They connect task composition to possible morphology; the next
 240 comparison asks which credit information helps a compatible tree learn.

241 All credit rules used the same exact local eligibilities and scalar output error. We compared exact paths with
 242 unit broadcast and a fixed six-site profile calibrated from each student’s mean path derivatives over 256 unlabeled
 243 initialization examples. Three development seeds selected learning rates with equal budgets; twenty fresh paired
 244 seeds tested the frozen choices. A separate nested target summed products over prefixes of two, four, six and eight
 245 inputs on its own compatible tree.

246 The initially calibrated profile supported pairwise learning but remained inaccurate on quartic and nested targets
 247 (Fig. 4B–D). At the primary 1,024-step checkpoint, mean normalized mean squared errors (NMSE) for exact and
 248 calibrated credit were 0.0317 and 0.0236 on pairwise targets, 0.1516 and 0.9813 on quartic targets, and 0.0230 and
 249 0.7690 on nested targets. NMSE divides squared prediction error by clean target variance. The predefined quartic-
 250 minus-pairwise difference in calibrated-minus-exact gaps was 0.838 (95% paired interval, 0.719–0.931), positive
 251 in all twenty seeds; a common learning rate gave 0.843 (0.727–0.935). Exact quartic learning was heterogeneous
 252 at this checkpoint: seventeen fits were near the noise floor and three remained stalled. The lower pairwise mean
 253 for calibrated credit likewise reflected one slow exact fit; exact credit performed better in fifteen of twenty seeds

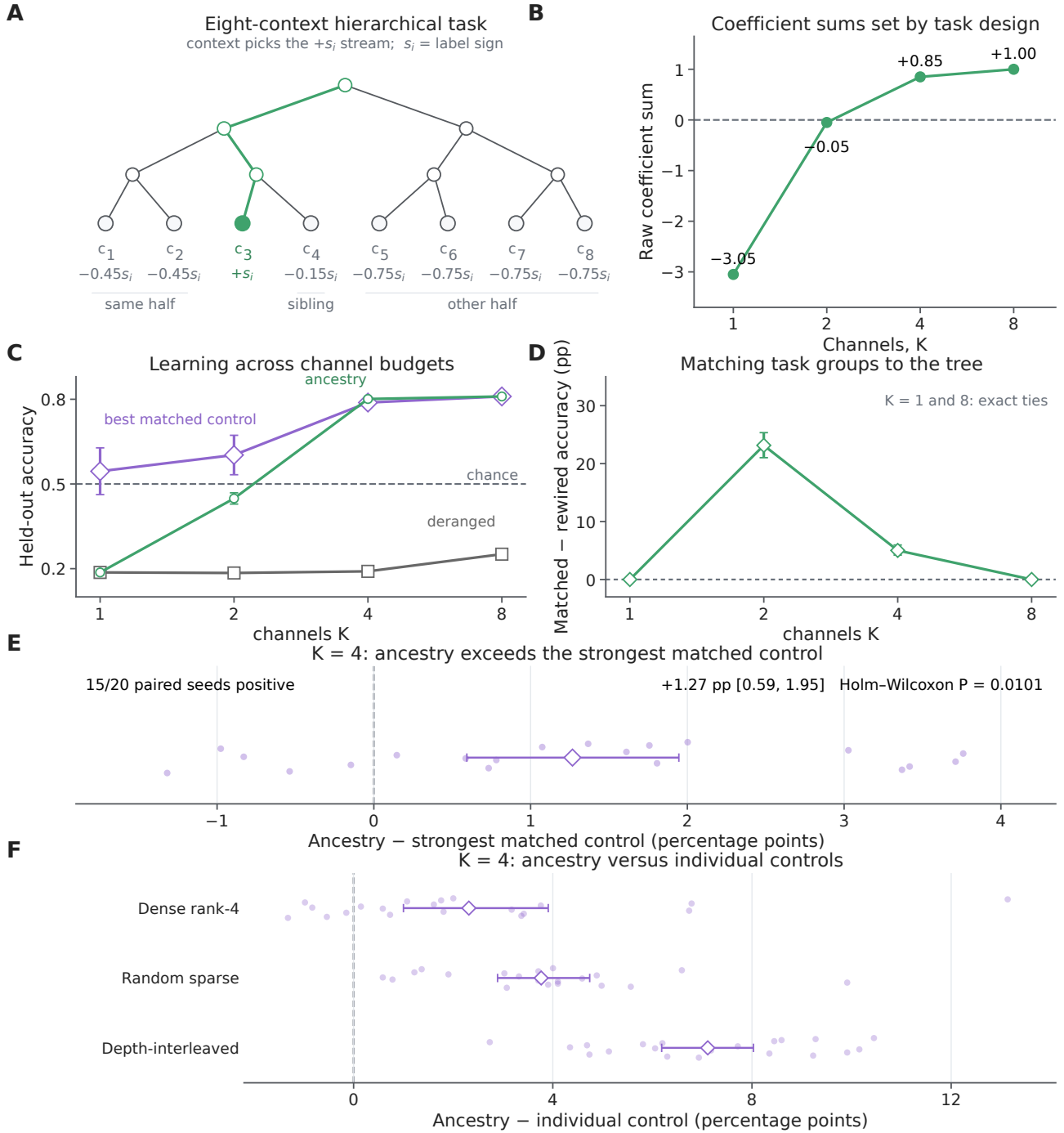


Figure 3: Hierarchical distractors favor ancestry at intermediate bandwidth. **A**, Eight streams carry label sign s_i along noisy teacher features. The selected stream has coefficient one; sibling, same-half and opposite-half distractors have coefficients -0.15 , -0.45 , -0.75 . **B**, Coefficient sums in the cued ancestry group at $K = 1, 2, 4, 8$. Implemented feedback divides these sums by square root of group size, preserving their signs. The four-route sign change follows from the design; it is not an independently discovered accuracy optimum. **C**, Held-out accuracy for correct ancestry, derangement and the per-seed maximum over four matched controls. This maximum is calculated after training, not learned by a selector. Dashed line: chance. **D**, Task-matched-minus-rewired accuracy at equal degree and depth. **E**, The four-route ancestry-minus-best-control effect, 1.27 pp (95% interval, 0.59–1.95), with 15/20 positive seeds and Holm-adjusted two-sided Wilcoxon $P = 0.0101$ across four budgets. **F**, Individual dense, sparse and depth-interleaved contrasts on a separate scale. The large derangement effect is shown in **C**. Symbols and bars are means and 95% seed-bootstrap intervals; small points in **E,F** are twenty paired seeds. Endpoint ties follow from coincident operators.

(Supplementary Figs. S15–S16).

Continuing every algebraic trajectory to 16,384 updates without retuning clarified the sustained comparison. All twenty selected-rate Adam exact quartic fits reached the noise floor by 4,096 updates, and the pairwise mean lag disappeared. At 16,384 updates, the quartic-minus-pairwise calibrated-credit deficit was 0.952 (95% paired interval, 0.883–1.012) at terminal states and 0.790 (0.691–0.880) under validation-based selection, positive in all twenty seeds (Fig. 4E). These continuations describe the same previously observed seeds under the same bounded dynamics: fixed-profile arms contacted coefficient bounds, whereas exact arms did not. Common-rate, per-seed, optimizer and bound controls are reported together in the Supplementary Information (Supplementary Figs. S15–S16 and Table S8).

The trained fields reveal a distinction missed by the input spectrum. Let $\mathbf{q}(x)$ contain the six nonsomatic derivatives of root output with respect to branch output; exact branch credit is scalar output error times $\mathbf{q}$. At the 1,024-step checkpoint, allowing an oracle amplitude per example, the best fixed profile captured 99.7% of exact-trained path-field energy for pairwise targets, but 46.1% for quartic and 44.8% for nested targets. The separation persisted at 16,384 updates: capture was approximately 100%, 39.6% and 40.5%, respectively (Fig. 4F). Loss-error weighting, eligibility weighting, site-normalized spectra and replay of 600 earlier fits preserved the original separation (Supplementary Section S4). Uniform-profile capture did not: a nearly one-dimensional field can have a signed, nonuniform direction. Positive fixed-profile scaling largely cancels in Adam, explaining calibrated/sign-only agreement; it does not explain the slower pairwise learning with changing exact paths.

Thus input-matched targets developed different credit spectra and tolerance of fixed profiles. At fixed weights and inputs, however, path derivatives are target-independent: their spectrum describes learned states and coordinates, not a task-only initialization diagnostic. Failure of the tested profiles supports example-dependent spatial credit without establishing that six independent channels or exact gradients are necessary.

Additional controls delimit the result. Fixed-profile Adam learned a compatible positive-conductance teacher accurately; broadcast learned aligned XOR-of-AND, whereas parity favored exact credit under the tested optimizer (Supplementary Figs. S13–S14 and S17). Noisy-query interaction estimation also yielded useful trees, retained as supporting structure-design evidence in Supplementary Fig. S35. Interaction structure therefore motivates specific credit comparisons rather than a universal monotonic resolution law.

Context-dependent branch tuning makes spatial credit useful in conductance trees

We next tested how branch tuning affects credit demand in a conductance tree. A first seven-compartment construction with monotonic branch responses was learned accurately by all tested rules. Under Adam, mean broadcast NMSE remained below 0.0006 after extended training despite a varying path field (Supplementary Fig. S18). We therefore tested a separate construction in which context selected branches with either aligned or opposing feature tuning (Fig. 5A,B).

Four terminal compartments fed two proximal compartments and a soma, using the directed E/I conductance balance (Methods). Both subtrees received the same positive feature pairs, $\exp(z_j)$ and $\exp(-z_j)$, for two independent latent inputs z_j . A binary context increased inhibitory conductance in one proximal compartment. Teacher excitation and inhibition produced increasing feature responses in both subtrees for the aligned task, but increasing responses in one and decreasing responses in the other for the opposed task. Teacher and student shared the circuit, ensuring representability. Students shared the teacher’s nominal proximal and coupling gains, with independent Gaussian perturbations of SD 0.4 in log conductance; terminal gains started near one. All 24 conductances were plastic, and paired conditions shared initial weights and example streams.

At fixed weights, binary context generates at most two exact path profiles. Their gains are positive: context changes the relative weighting of the two distal subtrees, while proximal credit is context invariant. The nominal parent’s conductance denominator changes by approximately eightfold. Thus branch selection is the relevant distinction, connecting this task to Fig. 2.

We tested whether that distinction requires oracle coefficients. A local rule delivered the somatic error to terminal parameters only when their parent was uninhibited, while retaining ungated error at proximal compartments and the soma. It used the supplied inhibitory context and exact local eligibilities, without evaluating exact paths or projecting

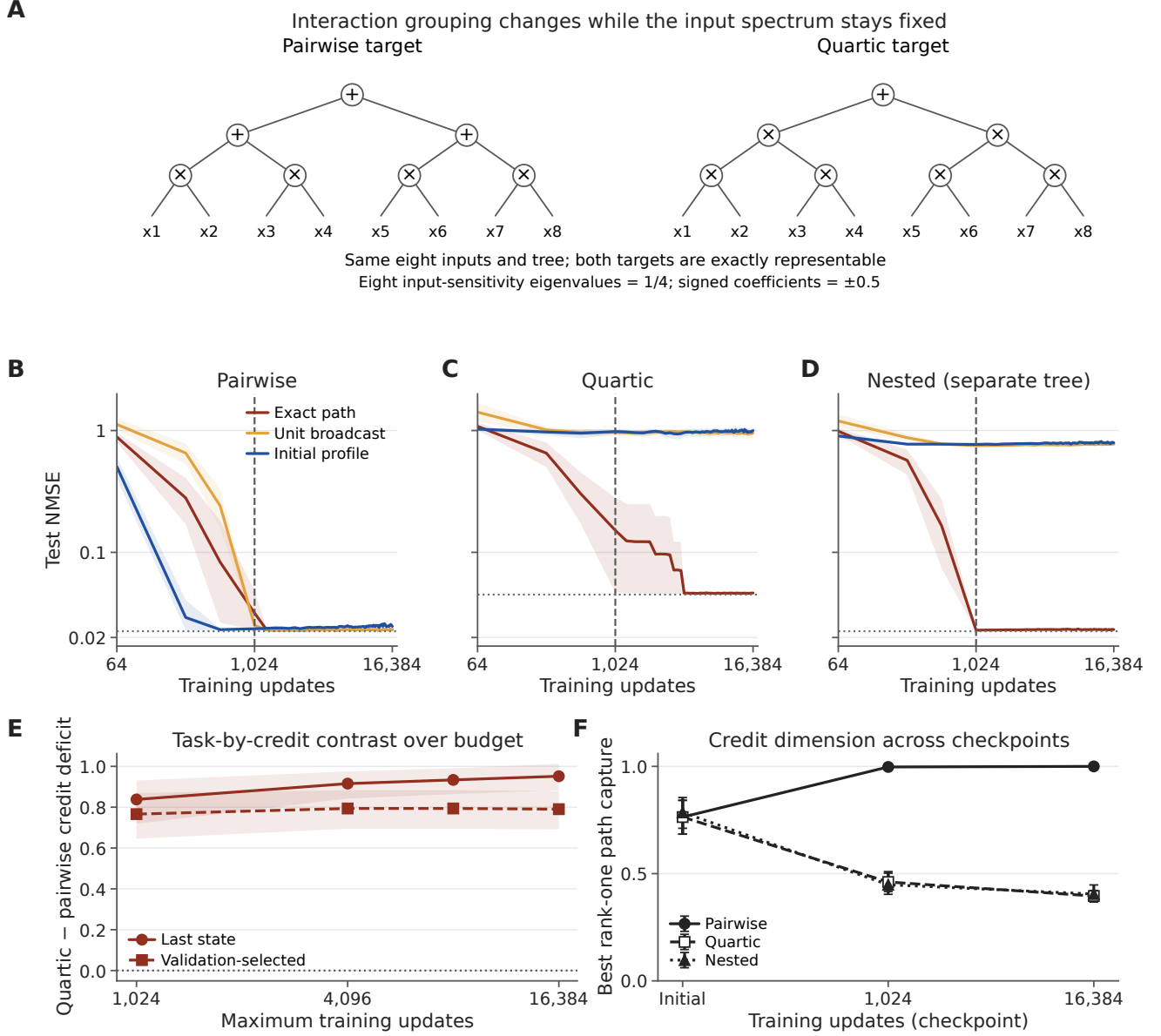


Figure 4: Matched input spectra yield different learned credit geometry and tolerance of fixed profiles. **A**, Pairwise and quartic targets share eight inputs, a compatible tree, initialization and examples. Signed coefficients have magnitude 0.5; both targets have input-sensitivity second moment $I_8/4$. Nodes illustrate interaction supports. **B–D**, Held-out normalized mean squared error (NMSE) for exact credit, unit broadcast and an initial profile calibrated on 256 unlabeled examples. Twenty seed blocks continue to 16,384 updates at the original development-selected Adam rates and coefficient bounds. Nested targets use a separate compatible tree. Curves are current-state means; shading shows pointwise descriptive 95% whole-seed intervals. Vertical lines mark the primary 1,024-step checkpoint; horizontal lines mark noise-only NMSE (0.0225 for pairwise/nested, 0.045 for quartic). **E**, Quartic-minus-pairwise difference in calibrated-minus-exact NMSE at current states and minimum-validation-error states within each budget, with paired 95% intervals. These continuations use previously observed seeds without retuning. **F**, Best rank-one capture of six-site exact path-field energy at initialization, 1,024 and 16,384 updates, with twenty-seed 95% intervals. The primary and continued checkpoints remain distinct; spatial directions and example-specific amplitudes are oracle fits.

an exact field. Two fixed distal profiles with oracle amplitudes and unit proximal credit supplied a separate capacity control. An additional proximal profile in the earlier three-profile control therefore addresses proximal updates, rather than another context distinction.

After exploratory panel results motivated this rule, we froze its protocol and tested twenty new paired seeds. All nine rules ran at three common rates through 16,384 updates; the primary rate and 4,096-update window were fixed before these outcomes. On opposed targets, mean held-out NMSE was 2.18×10^{-5} for exact credit, 2.33×10^{-5} for the local gate, 2.28×10^{-5} for the two-profile oracle with unit proximal credit, and 0.487 for broadcast (Fig. 5C,D). On aligned targets, the corresponding exact, local-gate and broadcast errors were 7.53×10^{-6} , 1.06×10^{-5} and 5.53×10^{-4} . The opposed-minus-aligned difference in broadcast-minus-gate error was 0.487 (95% paired interval, 0.360–0.615), positive in all twenty seeds. It remained 0.468 (0.332–0.604) within the longer window (Fig. 5E).

The controls identify what the gate must preserve. Reversing its distal selection gave opposed-task NMSE 0.949. Applying it at proximal compartments as well gave 0.123 and prevented both proximal inhibitory gains from learning: their eligibility is nonzero only during inhibition, precisely when that gate closes (Fig. 5F). The proper local rule changed all 24 parameters and succeeded across the three tested rates. A continuous shunt-proportional distal gate also worked (Supplementary Fig. S20).

This success is consistent with coarsening the exact rule’s strong context-dependent gain ratio. Earlier same-state gradient diagnostics showed that broadcast mixes opposing context-conditioned updates, despite nonnegative per-example alignment with exact gradients (Supplementary Section S5 and Source Data). The result establishes a local implementation when inhibitory context and somatic error are supplied; it does not explain their biological origin. All comparisons remain finite-budget and bounded-parameter results. Exact and local-gate fits did not reach conductance bounds, whereas most opposed broadcast fits did; the earlier wide-bound and rate controls remain in Supplementary Fig. S19.

Task organization determines the benefit of serial dendritic computation

The interaction comparison holds a compatible forward tree fixed while changing credit. We next varied serial forward structure itself, asking whether successive compartments help remove unwanted multiplicative changes in input strength. Physical depth D_p counts forward stages, whereas feedback resolution counts distinctions in the backward field.

Each network contained 64 somata and eight nonsomatic compartments per soma. D1 [8], D2 [2,3] and D3 [2,1,2] distributed those compartments across one to three stages in the three- and four-tier comparisons while preserving contacts and trainable resources. The two-tier D2 tree used [4,1], also with eight compartments. Branching vectors are ordered from soma to tips. A resource-identical grouped-point control connected all eight modules directly to the soma; raw-additive trees removed divisive conductance normalization. Flexible point networks provided additional, approximately parameter-matched comparisons.

Distal excitation carried class information multiplied by nuisance gains. Inhibitory sensors reported those gains with alignment α : zero retained the sensor distribution but removed trial-specific information, and one reported the factor exactly. With inhibitory reversal zero, the sensors entered the conductance denominator and could support successive divisions [24, 41]. Nested factors acted over fine, coarse and global input groups; flat factors had equal-resolution supports; locally cancellable factors placed the needed ratio within each module (Fig. 6A). The nested and flat targets involved commutative products. Their distinction was the spatial availability of factor information, rather than a different arithmetic order of multiplication.

In the 180-epoch hierarchy comparison, the two-tier task favored D2, whereas three- and four-tier tasks favored D3 among the tested architectures (Fig. 6B). The grid was incomplete, and its largest tested means identify budget-specific outcomes rather than a rule equating task tiers with optimal depth. Task-family controls showed what serial organization contributed: at full sensor alignment, exact-BP serial trees exceeded the resource-identical grouped-point control by 30.32 percentage points for nested gains and 22.20 points for flat gains, but had no advantage for locally available ratios. Removing trial-specific sensor information eliminated these large benefits (Fig. 6C). Serial structure was useful when the task required integrating factor information distributed across modules.

Credit comparisons within this useful forward architecture depended on training duration. Ten paired seeds

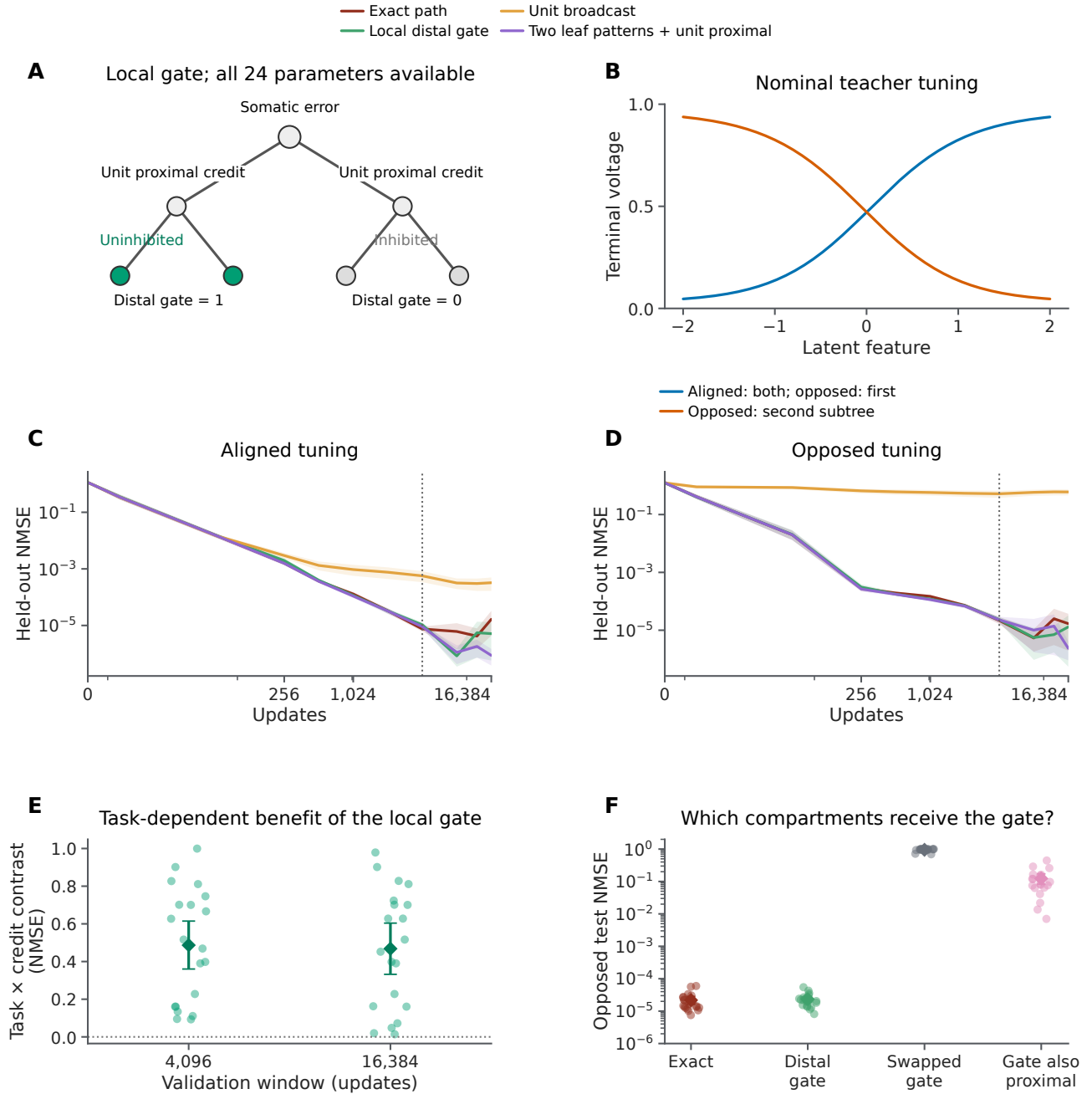


Figure 5: A local inhibitory gate supplies useful branch selection in a conductance tree. **A**, Four terminal compartments feed two proximal compartments and a soma. All 24 conductances are plastic. The local rule gates terminal credit by the parent's inhibitory activity while retaining unit multipliers for proximal and somatic credit. Local eligibilities are unchanged. **B**, Nominal teacher tuning for aligned and opposed branches; both tasks receive identical positive feature pairs. **C,D**, Fixed-checkpoint test NMSE for aligned and opposed targets, respectively. Twenty new paired seeds share initial students and example streams across rules. Adam rate 0.03 and both budgets were fixed before execution. The two-profile control uses oracle distal amplitudes and unit proximal credit. Dotted lines mark 4,096 updates; every trajectory continues to 16,384. Update axes are linear through 64 and logarithmic thereafter. **E**, Opposed-minus-aligned difference in broadcast-minus-local-gate NMSE, evaluated at minimum-validation-error states within each budget. **F**, Opposed-task outcomes within the primary window for correct, swapped and proximal-gated rules. Gating proximal credit freezes two inhibitory gains. Points show every seed; larger symbols and bands in **C–F** show means and descriptive 95% whole-seed bootstrap intervals. All rates and nine rules are retained in Supplementary Fig. S20. Earlier exploratory gate runs are excluded.

350 compared D1 exact BP with five D3 credit/optimization conditions for up to 600 epochs, retaining the original
 351 rates, parameter groups and thirty-epoch validation patience. These were matched restarts; the agreement with
 352 original-budget fits and complete validation histories are reported in SI. At validation-selected states from the same
 353 trajectories, exact-BP D3 exceeded D1 by 30.82 points at 180 epochs and 38.17 at 600 (Fig. 6D). Eight of ten
 354 D1 references stopped early; all fifty D3 fits reached the cap with validation loss still falling. These comparisons
 355 establish a forward benefit under the tested budgets, without equating optimization maturity across depths.

356 Within D3, the LocalCA exact-path-minus-shared-soma accuracy gap changed from +10.86 points at 180 epochs
 357 to −1.52 at 600 (95% paired interval, −1.83 to −1.23), with this budget-indexed sign change in all ten seeds. The
 358 mean gap remained negative from epoch 315, reached −2.94 points at epoch 486, and moved back towards zero by
 359 600 (Fig. 6E). Exact-path LocalCA nevertheless retained lower mean cross-entropy over epochs 180–600, ending at
 360 0.254 versus 0.276 (Fig. 6F). Accuracy counts threshold crossings, whereas cross-entropy also reflects predicted
 361 probabilities. Under the BP recipe, the 600-epoch exact-minus-broadcast accuracy gap was 0.48 points (0.40–0.55).
 362 Thus serial organization and spatial credit answer distinct questions: distributed factor information supports the
 363 forward architecture, while the eventual ranking of the tested credit rules remains unresolved (Supplementary
 364 Figs. S21–S23 and Table S9).

365 **Reconstructed arbors provide spatial routes beyond a shared broadcast**

366 MICrONS links reconstructed anatomy and connectivity to visual responses [42]. We used observed skeletons,
 367 soma positions and mapped contact locations and areas to model conductances, voltages and gradient fields, asking
 368 whether measured geometry supplies useful delivery patterns. This tests anatomical capacity rather than observed
 369 teaching signals.

370 Each inhibitory-bearing segment supplied a candidate route over its input-bearing descendants (Fig. 7A). Writing
 371 these supports as columns connects a branching arbor to a spatial dictionary; an additional constant column supplies
 372 a common broadcast (Fig. 7B). For a site-by-route matrix A_K , diagonal excitatory-area weights W and weighted
 373 least-squares projector $P_{A_K, W}$, capture of a field $\mathbf{t}$ was

$$\mathcal{C}(A_K; \mathbf{t}) = \frac{\|P_{A_K, W} \mathbf{t}\|_W^2}{\|\mathbf{t}\|_W^2}, \quad \|\mathbf{v}\|_W^2 = \mathbf{v}^T W \mathbf{v}. \quad (7)$$

374 For an ensemble of perturbations, numerator and denominator sum over field columns. A common-constrained SVD
 375 dictionary provided a ceiling at the same column budget. Controls used random anatomical routes, cable-distance
 376 bins, site-shuffled routes preserving nonzero counts, and surrogate trees preserving degree and depth while changing
 377 ancestry.

378 The fields were gradient changes induced by small focal shunts in reciprocal cables, generated without inspecting
 379 the selected dictionary. They nevertheless favor ancestry by construction: passive transport under a somatic loss has
 380 the ancestry structure derived below. This tests how well a small dictionary compresses the tree’s own perturbation
 381 responses. In the initial eight cells, broadcast captured 31.4% of field energy, near the 32.6% rank-one ceiling. Every
 382 dictionary therefore contained one common broadcast and $K - 1$ spatial profiles, with the protocol fixed before
 383 evaluation in 47 disjoint cells from the same mouse and a reconstructed second mouse.

384 Varying K separates the common signal from the additional spatial information each dictionary can capture
 385 (Fig. 7C). At eight columns in the 47-cell cohort, broadcast-plus-ancestry captured 0.619 of total energy versus 0.578
 386 for degree–depth surrogates. The paired advantage was 0.041 (95% cell-bootstrap interval, 0.020–0.062), positive
 387 in 38 of 47 cells. This mean conceals heterogeneity: in 11 cells, more than half of the 200 surrogates matched
 388 or exceeded the actual tree. Depth bins captured 0.511 and site-shuffled profiles 0.436; the ancestry-minus-depth
 389 difference was 0.108 (0.077–0.135), positive in 45 cells. After removing broadcast, ancestry and depth-bin residual
 390 capture was 0.527 and 0.392. The paired residual contrasts show the spatial advantage separately from the shared
 391 component (Fig. 7D). Actual ancestry improves average compression within this passive response family, while
 392 coarse geometry explains much of the capacity.

393 Capture and delivery cost are separate constraints (Fig. 7E). The eight-column ancestry dictionary used 18.8% of
 394 the entries of a fully dense dictionary on average; its site-shuffled control preserved exactly the same nonzero count.

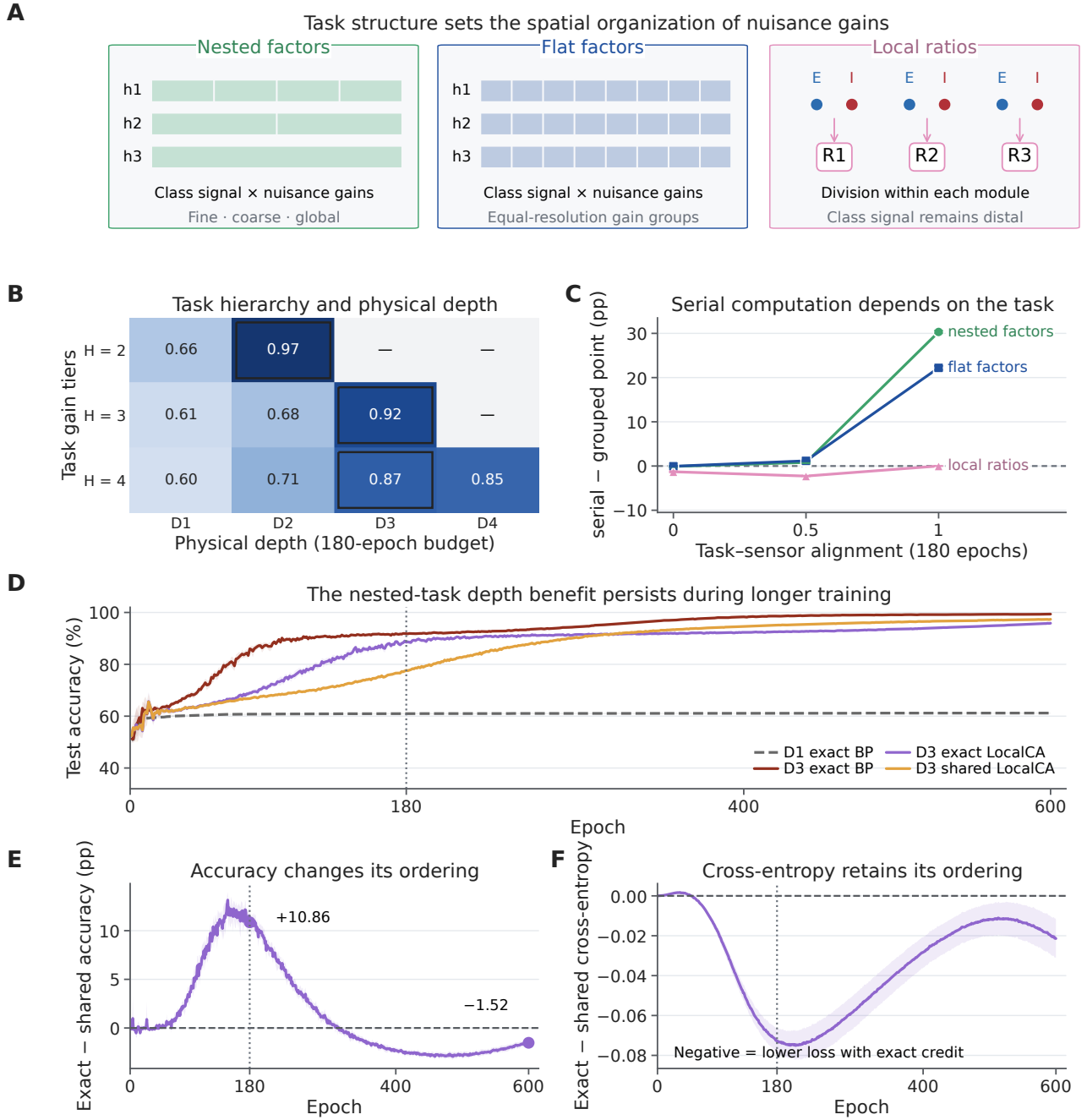


Figure 6: Task organization determines when serial dendritic computation helps. **A**, Nested gains have fine, coarse and global supports; flat gains have equal-resolution supports; local-ratio tasks expose division within each module. Drawings indicate input access, not exact cancellation. E/I, excitatory/inhibitory inputs. **B**, Mean exact-BP accuracy from ten seeds per tested task gain-tier H and physical-depth D combination at 180 epochs. Outlines mark the largest tested means; dashes denote untested combinations. The incomplete grid does not establish converged depth optima. **C**, Exact-BP serial-minus-resource-identical-grouped-point accuracy across task families and sensor alignment, with ten paired seeds and 95% intervals at 180 epochs. **D**, Nested-task D1 [8] and D3 [2, 1, 2] trajectories: 64 somata, eight nonsomatic compartments per soma and matched resources. Validation-selected states are carried forward after stopping. **E,F**, Paired D3 exact-minus-shared LocalCA differences in accuracy and cross-entropy. Accuracy changes sign between 180 and 600 epochs; exact credit retains lower mean cross-entropy. All fifty D3 fits reach the 600-epoch cap with falling validation loss; eight of ten D1 references stop earlier. Shading in **D–F** shows pointwise descriptive 95% intervals over ten paired seeds. Both recipes use Adam with their original rates and parameter groups; validation selection does not establish convergence.

Equal column count does not ensure equal rank, and nonzero entries are a delivery-cost proxy rather than feedback synapses or metabolic expenditure.

The initial eight cells retained the same ordering (Fig. 7F). In the Pinky second mouse, eight of ten qualifying cells supported the eight-column comparison. Their 9–13 input-bearing sites put ancestry near the residual-capture ceiling, so lower-budget comparisons and exclusions remain separate from the larger arbors. Direct E/I labels covered 4.71% of disjoint-cohort inputs; the initial cohort’s 73.0% direct-versus-proxy agreement concerned jointly labeled contacts. Geometry, labeling and compression sensitivities accompany the cohort controls in SI (Supplementary Figs. S24–S27).

Focal shunting sets gains on ancestry-defined credit routes

Morphology supplies spatial routes; conductance can regulate their amplitudes. We examined this connection by comparing a focal inhibitory shunt with a current injection matching its baseline focal current in reconstructed reciprocal cables. A separately solved somatic current restored baseline somatic voltage after either intervention. The endpoint was the change in the modeled excitatory-conductance gradient.

Let G be the passive conductance matrix and $\mathbf{b}$ the current-source vector, so that baseline voltages satisfy $G\mathbf{V} = \mathbf{b}$. With $\mathbf{e}_k$ the unit vector at compartment k , adding inhibitory conductance κ_k at compartment k changes both G and its source: $G' = G + \kappa_k \mathbf{e}_k \mathbf{e}_k^\top$, $\mathbf{b}' = \mathbf{b} + \kappa_k E_1 \mathbf{e}_k$. The matched injection supplies baseline current $\kappa_k(E_1 - V_k)$ without changing G . The adjoint $\mathbf{q}$ gives loss sensitivity to current injected at each compartment. For a fixed somatic loss derivative, it changes by

$$\mathbf{q}' = \mathbf{q} - \frac{\kappa_k q_k}{1 + \kappa_k (G^{-1})_{kk}} G^{-1} \mathbf{e}_k. \quad (8)$$

The column $G^{-1} \mathbf{e}_k$ gives the voltage response to unit current at k . On a passive tree with a somatic loss, the resulting adjoint change has an exact ancestry structure: all descendants of the shunt share the attenuation factor $[1 + \kappa_k (G^{-1})_{kk}]^{-1}$, while each sister block along the path to the soma shares its own factor (Fig. 8A; Supplementary Section S8). These disjoint blocks form an ancestry partition. Weighting their indicators by the baseline soma-to-site transfer gives a dictionary on which the shunt acts through diagonal gains, providing a physical instance of a diagonal channel-gain matrix, denoted Γ_u . The diagonal form applies in this partition basis; it need not hold for an arbitrary selected set of overlapping routes.

In the small-perturbation fields used for Fig. 7, the driving-force term was substantial: its weighted squared energy ranged from 0.99 to 3.50 times that of the full field across the eight initial cells. It opposed the adjoint term (weighted correlations -0.979 to -0.913). Despite this cancellation, 97.1–99.9% of full-field energy remained within each focal ancestry partition. The small outside-partition residual therefore does not imply a negligible driving-force contribution (Supplementary Section S8).

We compared descendants $D(k)$ with off-route sites $C(k)$ at similar depth. A separate kernel criterion tests whether the adjoint change is larger below the shunt (Methods); within-cell, reversal-potential and contact-label controls are in Supplementary Fig. S28B,C. The analyzed gradient was sensitivity to open excitatory conductance, $\gamma_i = q_i(E_E - V_i)$. The local driving force $E_E - V_i$ can change differently within each block, so the complete synaptic gradient need not share a single block gain. With primes denoting post-intervention values and ε_γ a numerical floor defined in Methods, we measured absolute log change, $m_i = |\log[(|\gamma'_i| + \varepsilon_\gamma)/(|\gamma_i| + \varepsilon_\gamma)]|$, and localization $\Lambda_k = \text{median}_{D(k)} m_i - \text{median}_{C(k)} m_i$. Positive localization means a larger gradient change below the shunt.

In the permissive normalized model, shunts changed descendant gradients more strongly than depth-matched off-route gradients, with increasing localization at higher dose (Fig. 8B; Supplementary Fig. S28D). Current injection changed descendant gradients roughly threefold less. Its off-route changes were zero by construction under the soma-restoring control; this is not a general claim that current injection cannot change a gradient. Holding the post-shunt voltage but replacing its adjoint with the baseline adjoint removed most localization, supporting a transport contribution beyond the voltage change alone (Fig. 8C).

Physical calibration constrained the spatial selectivity of the gradient change. At axial resistivity $R_a = 150 \Omega \text{cm}$ and specific membrane resistance $R_m = 15,000 \Omega \text{cm}^2$, mean shunt-minus-current localization was small and opposite-signed in the initial eight cells and a disjoint 45-cell cohort from the same mouse: -0.0019 and $+0.0024$, respectively.

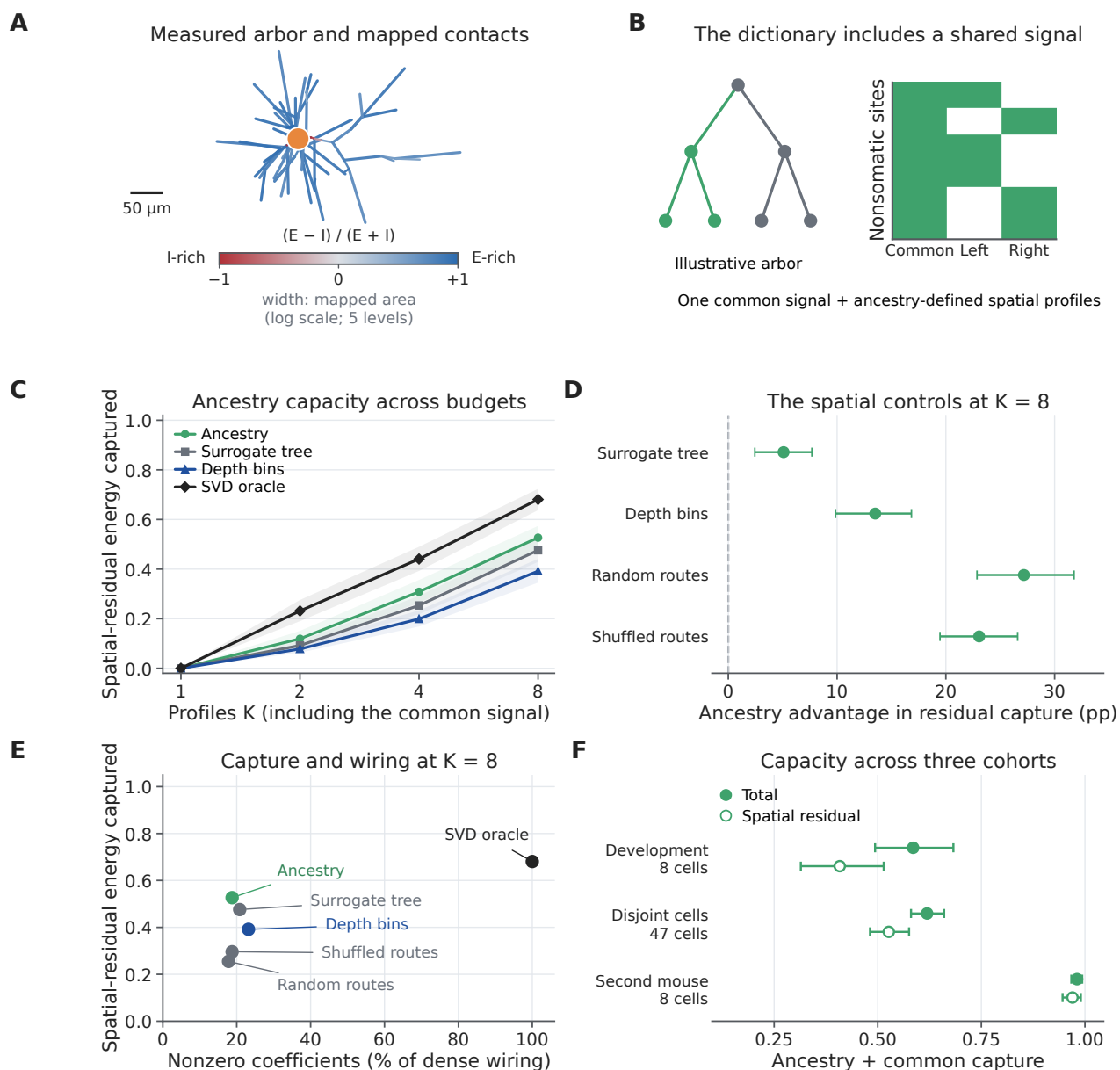


Figure 7: Anatomy supplies sparse spatial dictionaries beyond a shared broadcast. **A**, Median-sized arbor from the initial eight-cell cohort. Color represents mapped E/I contact-area balance; width represents total contact area. Scale bar, 50 μ m. **B**, Illustrative subtree-to-column mapping with a shared constant profile. **C**, Capture of field energy remaining after a weighted common projection at $K = 1, 2, 4, 8$, each evaluated in all 47 disjoint v661 cells. Every dictionary contains the same constant profile; the common-constrained SVD is an oracle ceiling. **D**, Paired ancestry advantages at $K = 8$. Surrogate trees preserve segment depths and parent out-degrees; shuffling preserves column nonzero counts. Randomized controls average 200 draws per cell. **E**, Residual capture versus mean nonzero coefficient density, relative to dense eight-column wiring. This is a delivery-cost proxy; equal K does not equate rank or wiring. **F**, Total and residual ancestry capture in the initial, disjoint and second-mouse cohorts. Pinky's eight eligible cells have only 9–13 input-bearing sites. Symbols and intervals in **C–F** summarize cell means and descriptive 95% cell-bootstrap intervals where shown. Capture uses squared excitatory-area-weighted energy. Passive shunt fields inherit ancestry structure through cable physics; this is anatomical compression capacity, not observed teaching.

441 Signed changes showed attenuation of both descendant and off-route gradients at high resistance; lowering resistance
442 primarily increased descendant attenuation (Fig. 8D). This increased the localization contrast by more than an order
443 of magnitude (Fig. 8E). Distributed background conductance produced the same qualitative recovery at fixed high
444 membrane resistance (Supplementary Section S8 and Source Data), consistent with a dependence on electrotonic
445 state [25, 43].

446 Adding weak voltage-dependent channel currents produced a dose response close to the passive response at
447 $R_m = 1,000 \Omega \text{cm}^2$ with one baseline-leak unit of distributed background conductance (Supplementary Section S8;
448 Fig. S29D,E). This checks the local linearization around solved steady states. The baseline Jacobian included gate
449 derivatives but remained fixed across doses, so the check does not establish robustness to strongly active dendrites or
450 channel kinetics.

451 Together, the exact partition identity and the electrical calibration connect the morphology-defined dictionary
452 to conductance-dependent route gains [26]. Their spatial selectivity depends on cable state. These imposed, state-
453 matched perturbations measure modeled transport, rather than endogenous error production, observed plasticity or a
454 learning advantage.

455 Measured responses bound the anatomical alignment proposal

456 The anatomical dictionary and shunt calculations establish capacity and transport. Measured responses ask whether
457 functional relationships preferentially follow ancestry. Seven MICrONS targets had complete arbors and imaged
458 partners: 69 partners in the selected-scan dataset and thirteen eligible scans in the scan-complete analysis. These
459 recordings sampled stimulus responses rather than learning.

460 Partners with more shared dendritic ancestry did not have reliably more similar condition-mean responses after
461 controlling Euclidean separation and soma-to-contact path-length differences. Mean partial rank correlations were
462 -0.069 for selected scans and -0.048 for all eligible scans; both target-bootstrap intervals included zero (Fig. 9A).
463 This test therefore provides no evidence of preferential ancestry alignment in the sampled partners. Its scale differs
464 from population-level like-to-like wiring [44].

465 To establish the sensitivity of this small sample, response-level simulations preserved observed partners, shared
466 inputs, repeated stimuli and seven-target inference, with noise calibrated to measured repeat reliabilities. Under
467 the specified Gaussian ancestry-signal model, 80% detection required a mean simulated target-level partial rank
468 correlation of approximately 0.25 (Fig. 9B). The observed repeat reliabilities used for attenuation vary across
469 recordings (Fig. 9C). This conditional threshold leaves weaker effects, different signal geometries and unobserved
470 partners unresolved (Methods; Supplementary Section S9).

471 A separate passive-tree model predicted each target's responses from its recorded partners. Its selected ancestry
472 routes reached only 48.0% of mapped inputs, while one unrestricted fixed transfer profile already reconstructed
473 98.4% of exact credit and 97.6% of exact updates. Restricting that profile to incomplete route supports therefore
474 tests transfer geometry more than example-dependent credit. Mean held-out NMSE was 0.788 for exact learning,
475 0.803 for ridge regression and 0.832 for the ancestry rule; the ancestry-minus-ridge difference was uncertain (0.029;
476 95% target-bootstrap interval, -0.013 to 0.074). Allowing oracle coefficients within the same ancestry span barely
477 improved update reconstruction. Full support, prediction, input-removal and reconstruction controls are reported
478 together in the Supplementary Information (Supplementary Figs. S31–S32). This model diagnostic and the measured
479 ancestry-similarity test constrain different parts of the anatomical proposal.

480 Discussion

481 Dendritic trees provide spatial addresses through which learning can distinguish conflicting synaptic demands. Which
482 distinctions are useful depends on the task and the local sensitivity already available to the learner. Morphology
483 supplies delivery patterns, teaching mechanisms supply their coefficients, and conductance regulates their gains.
484 Image learning benefits mainly from preserving neuronal identity, including with random between-neuron feedback.
485 Contextual conflict favors branch selection, supplied through feedback or local eligibility gating. Hierarchical

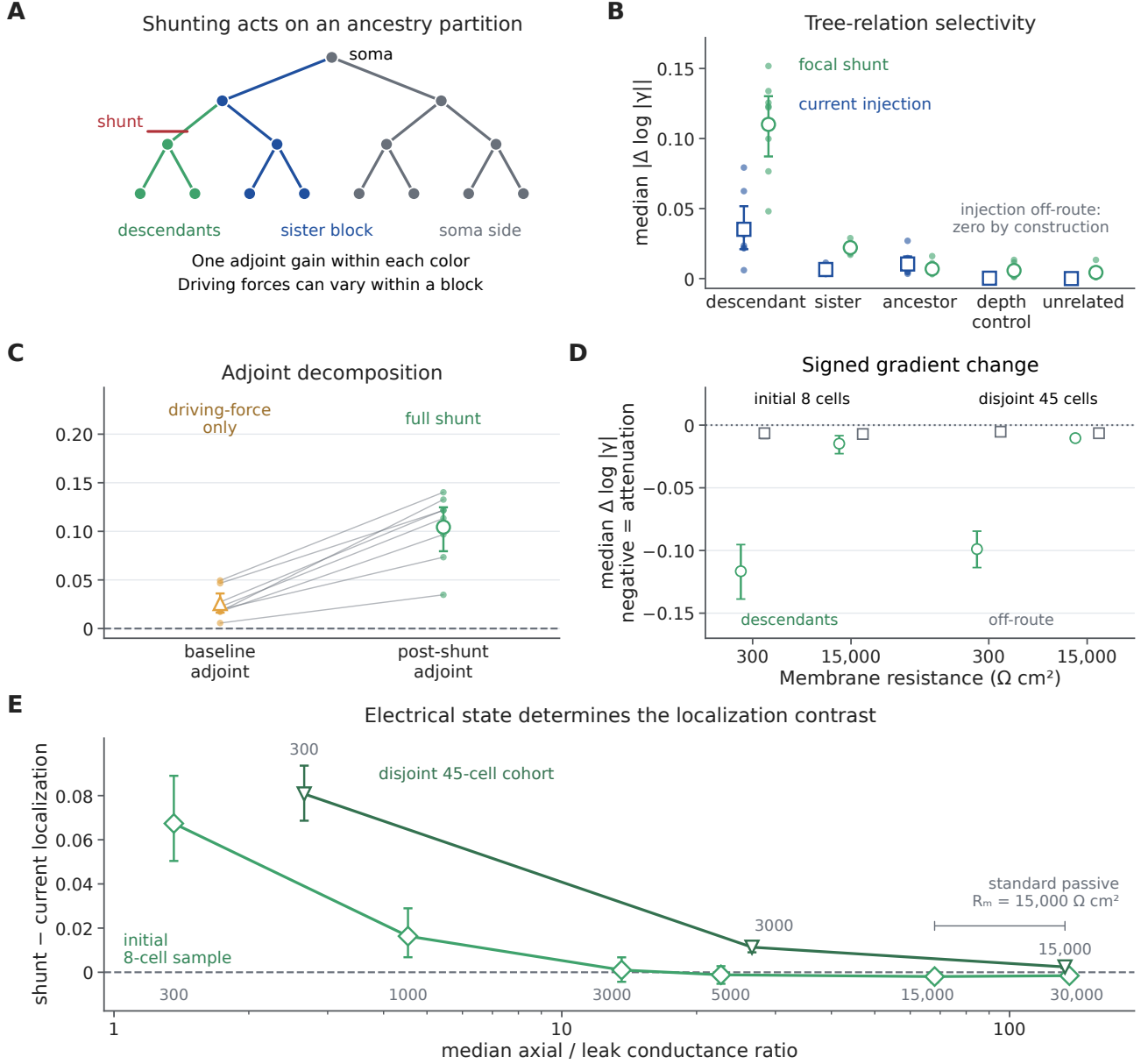


Figure 8: Focal shunts set ancestry-defined gains whose selectivity depends on electrical state. **A**, On a passive tree with a fixed somatic loss derivative, a shunt gives the same adjoint attenuation factor to every site in each colored ancestry block. The resulting gain is diagonal in the baseline-weighted partition dictionary; local synaptic driving forces can still vary within a block. **B**, Absolute log change in excitatory-conductance gradient magnitude by tree relation at unit dose, in eight reconstructed cells under permissive normalized passive parameters. A current injection matches baseline focal current, and a separately solved somatic current restores baseline somatic voltage after either intervention. Off-route injection zeros arise from this construction. **C**, Keeping the post-shunt voltage while restoring the baseline adjoint removes most localization. Lines pair cells. **D**, Signed log change in gradient magnitude at the two physical-calibration endpoints, with $R_a = 150 \Omega \text{ cm}$ and $R_m = 300$ or $15,000 \Omega \text{ cm}^2$. Circles and squares show descendants and depth-matched off-route sites; negative values denote attenuation. Site medians are averaged within cells; symbols and whiskers are cohort means and 95% whole-cell bootstrap intervals (20,000 draws). **E**, Electrical calibration in the initial eight cells and a disjoint 45-cell cohort from the same mouse. The horizontal axis is the cohort-median axial/leak ratio at fixed axial resistivity $R_a = 150 \Omega \text{ cm}$; numerical labels give membrane resistance R_m in $\Omega \text{ cm}^2$. Shunt-minus-current contrasts are small and opposite-signed at $R_m = 15,000$, and increase at lower resistance. Symbols and whiskers are cohort means and 95% cell-bootstrap intervals after averaging sites within cell. These state-matched perturbations measure modeled transport, not plasticity or learning benefit.

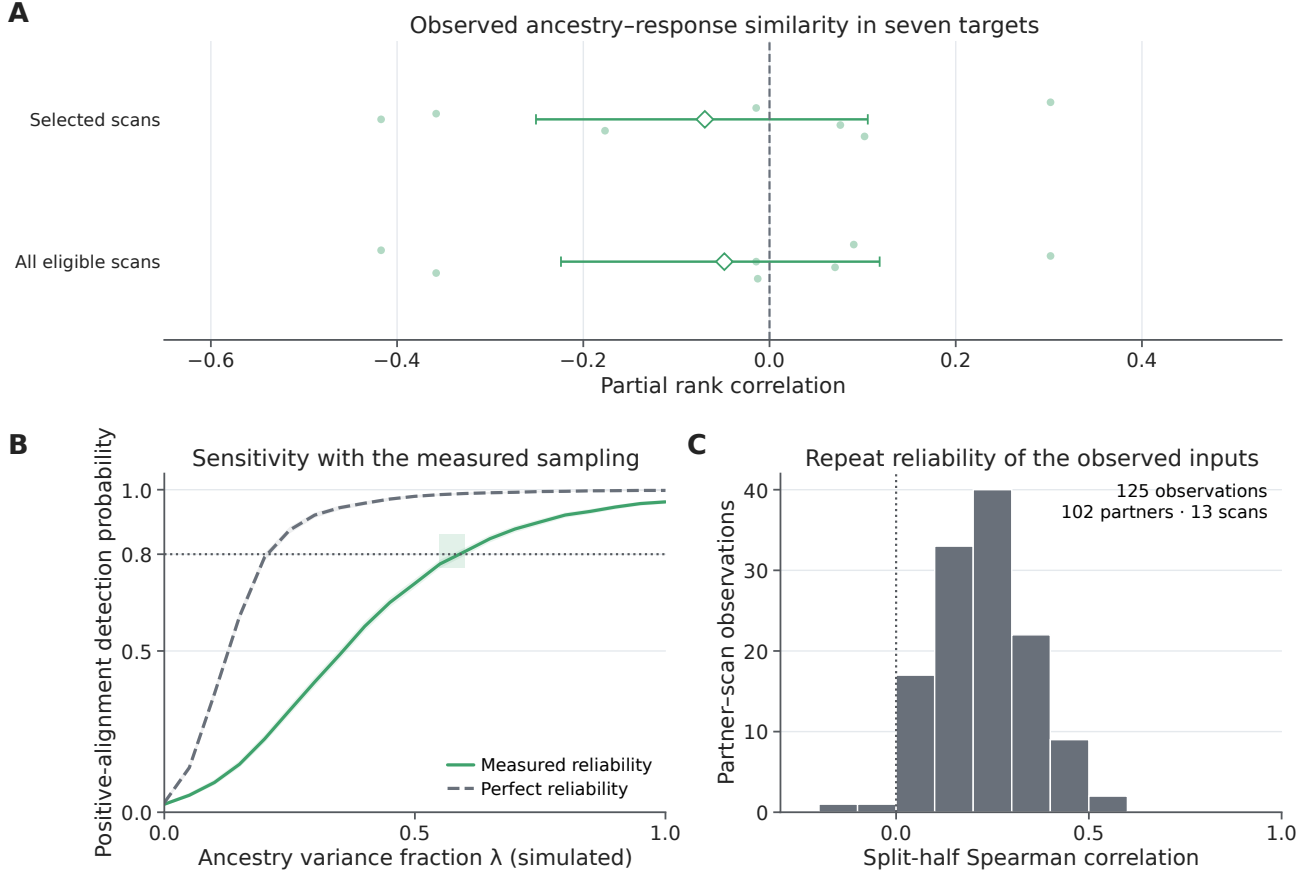


Figure 9: Measured ancestry alignment and the sensitivity of the available sample. **A**, Ancestry–response partial rank correlations after controlling Euclidean separation and contact-depth differences. Points show seven targets; diamonds and intervals show means and 95% target-bootstrap intervals. Selected scans contain 69 partners; all eligible scans comprise thirteen scans of the same targets. **B**, Conditional positive-alignment detection probability under measured or perfect response reliability. The simulated fraction λ allocates latent response variance to a Gaussian ancestry kernel; it is not an observed correlation. Each point summarizes 4,000 datasets analyzed by the exact two-sided seven-target signed-rank test at $P \leq 0.05$, also requiring a positive mean. Shading shows 95% Monte Carlo intervals. With measured reliability, the 80% threshold lies between $\lambda = 0.55$ and 0.60, corresponding to mean partial correlation approximately 0.25 under this model. **C**, Measured split-half response correlations: 125 partner–scan records, 102 unique partners and thirteen scans. Negative values are retained here and clipped to zero only when calibrating nonnegative signal variance in the simulation. Support geometry, ridge comparisons and update reconstruction are reported together in SI.

distractors favor ancestry at a bandwidth predicted by the generator's coefficients. These results distinguish useful information from its physical carrier.

The interaction comparison shows why input rank alone is insufficient. Pairwise and quartic targets share input-sensitivity spectra and a compatible tree but develop different credit spectra and tolerance of fixed profiles, despite initial calibration and common-rate controls. Spatial direction matters alongside dimension: even a nearly one-dimensional field can be poorly captured by uniform broadcast. These are learned-state properties, not retrospective support for the failed initialization selector. Intermediate dictionaries, adaptive profiles and alternative dynamics remain plausible solutions.

Conductance trees connect this distinction to a dendritic computation. Simple credit learns some compatible teachers accurately, whereas opposing branch tuning under contextual inhibition benefits from branch selection. A local inhibitory gate supplies that selection without oracle path coefficients, provided proximal credit remains ungated. The task supplies the inhibitory context and somatic error; discovering these signals remains a separate problem. Forward compatibility remains essential: the scalar multi-affine cut criterion connects input grouping to hierarchical tensor descriptions and supports structure estimation from noisy examples [45–47]. Serial composition likewise helps remove distributed nuisance factors, but the LocalCA accuracy advantage of exact credit changes sign between the 180- and 600-epoch budgets while exact credit retains lower cross-entropy. The gap is still moving at the cap. Task rank, physical depth and credit resolution therefore describe different constraints.

Obtaining coefficients remains a limitation beyond this explicit context gate. In the separate soft-encoder experiment, the prespecified noisy-cue condition remained 54.47 percentage points below oracle routing (Supplementary Fig. S10D and Section S3). Even with sixteen noise-free calibration trials, the encoder identified every context yet achieved only 19.4% task accuracy; exploratory maximum-probability readout reached the 80.3% oracle accuracy. This isolates leakage into incompatible routes as a failure mechanism. The exploratory rescue qualifies the primary deficit without replacing it. Exact paths and projected coefficients elsewhere remain information-access controls, and successful routing requires both an adequate span and coefficients that preserve its useful distinctions.

Reconstructed arbors compress their modeled passive response fields beyond a common broadcast. Degree–depth surrogates explain much of this capacity, and the target fields themselves favor ancestry through passive transport under a somatic loss. Testing alignment beyond that construction requires measured fields or a model with different error sources or active dynamics; changing the passive perturbation family alone would not suffice. Focal shunting supplies a biophysical link by reweighting the baseline-weighted ancestry partition. Driving-force changes partly cancel adjoint changes, while cable state sets the spatial selectivity of the resulting gradient. This concerns sensitivity transport, distinct from inhibition that computes errors or regulates forward gain and load [48–51].

Measured ancestry–response similarity provided no evidence of preferential alignment in the small recorded cohort. The offline learning comparison chiefly tests preservation of an almost fixed transfer profile and offers limited evidence about endogenous route use. A direct experiment would compare plasticity below a focal inhibitory site with electrotonically matched off-route sites while matching somatic voltage, first-order current and outcome signal. Changing conductance state would test the predicted separation between shared and ancestry-selective effects [22, 52, 53].

The framework complements somatic prediction, segregated teaching, interneuron-mediated errors and burst plasticity [32, 34–36], alongside cable-learning and engineered gating models [39, 54–57]. Spikes, temporal eligibility, recurrence and channel kinetics may change the available profiles and coefficient estimation [58, 59]. The organizing question remains which task-relevant distinctions the arbor and learning mechanism can jointly preserve.

Methods

The Methods define the models, task manipulations and inferential units behind the main comparisons. The Supplementary Information's contents and cohort index (Table S11) locate derivations and complete protocols. Section S1 covers exact and restricted credit; S2–S3 cover images, conflict and ancestry; S4–S6 cover interactions, conductance learning and physical depth; S7–S9 cover anatomy, shunting and measured responses. S10 presents optional structure-selection studies, statistics and reproducibility. Resolved configurations, seeds, checkpoints,

exclusions and source hashes accompany the numerical tables. Reusing a cohort for a diagnostic does not add independent observations.

Conductance models and delivery coordinates

In a directed tree, $[b_1, \dots, b_D]$ lists branching factors from soma to tips, giving $\sum_{d=1}^D \prod_{r=1}^d b_r$ nonsomatic compartments. A child c supplies output $a_c = f_c(V_c)$. With leak g_n^L , aggregate open excitatory/inhibitory conductances $G_n^X = \sum_{i \in \mathcal{I}_n^X} g_i^X x_i^X$ for contact class X , and child couplings $g_{c \rightarrow n}^{\text{den}}$, the steady voltage is

$$V_n = R_n^{\text{tot}} \left(g_n^L E_L + G_n^E E_E + G_n^I E_I + \sum_c g_{c \rightarrow n}^{\text{den}} a_c \right), \quad (R_n^{\text{tot}})^{-1} = g_n^L + G_n^E + G_n^I + \sum_c g_{c \rightarrow n}^{\text{den}}. \quad (9)$$

Here $x_i \geq 0$ is contact activity, $g_i \geq 0$ its conductance, and E_X its reversal potential. A terminal-to-soma pass evaluates the directed steady state. Artificial conductance trees use dimensionless leak one and reversals $E_L = E_I = 0$, $E_E = 1$. Monotone parameter transforms enforce nonnegative conductances. Branch-output nonlinearities, termed reactivations in the implementation, are identity functions when disabled. Both excitatory and inhibitory contacts learn in the regular-tree experiments.

For a reciprocal cable residual $\mathbf{F}(\mathbf{V}, \mathbf{g}, \mathbf{x}) = 0$, the nonsingular voltage Jacobian J_V gives the adjoint

$$J_V^\top \mathbf{q} = \nabla_{\mathbf{V}} \mathcal{L}, \quad \partial \mathcal{L} / \partial g_i = -\mathbf{q}^\top \partial \mathbf{F} / \partial g_i. \quad (10)$$

With passive current balance, the synaptic gradient is $x_i(E_i^{\text{rev}} - V_n)q_n$. This established sensitivity calculation supplies the reference for restricted credit [9, 10, 33, 39]. Directed LocalCA combines local eligibility with delivered credit. For output $a_n = f_n(V_n)$, activation error $\delta_n^a = \partial \mathcal{L} / \partial a_n$ gives voltage error $\delta_n^V = f'_n(V_n) \delta_n^a$. The image implementation retains f'_n in local eligibility and replaces δ_n^a ; its gate updates also retain their exact local derivatives. Thus a fixed activation-space profile can correspond to a state-dependent voltage-space profile. Supplying the exact field recovers the exact selected gradients.

The raw-additive control retains the graph but sums signed inputs without the conductance denominator or reversal-potential driving force. Historical normalized-additive controls additionally standardize voltage across stage coordinates within each example; Supplementary Section S2 defines those variants. The optional four- and five-factor rules add stage-level preconditioners and are described in Supplementary Section S1.

Writing a delivery dictionary as $A\Gamma$ separates spatial profiles in A from diagonal channel gains in Γ . Nonzero gains preserve the column span and oracle capture, while changing delivered coefficients and conditioning. A passive focal shunt admits this representation in its baseline-weighted ancestry partition; arbitrary overlapping dictionaries need not. Physical stage count, feedback rank, external error dimension and state-dependent path resolution are distinct resources (Supplementary Tables S1–S2).

Image feedback ladders and checkpoint diagnostics

For example i , the trained linear decoder supplies activation errors $\delta_{i,u} = \partial \mathcal{L} / \partial a_{i,0,u}$ at the somatic outputs of neurons u . The strict scalar is their mean unless its magnitude is at most 10^{-6} , in which case it is the largest-magnitude coordinate, preserving that coordinate’s sign. The same scalar reaches every core compartment, including the somatic gate. Neuron-specific feedback repeats each neuron’s own coordinate throughout its arbor; exact transport multiplies it by the local path derivatives. The historical matched-width fallback retained neuronal identity at the 128-wide somatic stage, affecting 640 of 2,413,312 core parameters. A separately trained strict-scalar control tests that exception (Supplementary Section S2 and Source Data indexed in Table S12).

The new MNIST ladder uses 128-soma [3, 3] shunting and raw-additive networks, with 40 excitatory and 20 inhibitory contacts per nonsomatic compartment. Within each architecture, rules share initial contact construction, initialization policy, decoder and data streams. Pixels are divided by 255 and flattened without further normalization or augmentation. Each seed divides the 60,000 training images into 48,000 training and 12,000 validation examples; all 10,000 official test images remain held out. Training uses 180 epochs, batches of 256 and the minimum-validation-cross-entropy checkpoint, without early stopping. Adam uses $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ and no weight decay.

Original rates are 0.0015 for input synapses and decoder, 0.0007 for branch couplings and 0.0005 for reactivation gates.

Projected $K = 1$ averages all twelve nonsomatic activation errors. The $K = 3$ dictionary instead averages each proximal compartment with its three distal children. Both use the exact field and preserve its mean at identical weights; their norms can differ. Both projected rules retain exact somatic errors and local activation derivatives. A decoder-only reference trains the readout on the initialized core, whose unchanged hash is verified. Three development seeds compare rate multipliers 0.3, 1 and 3 for every rule and architecture, giving 108 fits. Validation alone selects rates. Ten fresh paired seeds per architecture and rule give 120 selected-rate and 120 original-common-rate outcomes from 190 unique fits; 50 fits serve both policies. The projected- $K = 1$ addendum was frozen before fresh outcomes. Complete protocols, clipping and excluded implementation checks are in Supplementary Section S2.

The original fifteen-seed MNIST ladder is retained in Supplementary Section S2 and Source Data; its gradient diagnostics appear in Fig. S7E–G. Fashion-MNIST uses ten paired seeds; the raw-additive CIFAR-10 ladder uses twenty and a 20-soma $[3, 3, 3, 3]$ tree. Their distinct protocols and initialization are retained in SI. The separate fifteen-seed direct-feedback-alignment factorial replaces the decoder derivative by a fixed Gaussian output-to-neuron matrix, scaled by inverse square root of output dimension. It pairs random-feedback fits with the original exact-readout reference. Ownership controls permute neuronal recipients without changing coordinate values or bandwidth.

Common-checkpoint diagnostics compare rules at identical weights and states. Exact path transport is checked against automatic differentiation. New capture measurements use 2,048 held-out images at initial and exact-trained states; activation- and voltage-space metrics are kept separate. Gradient cosine, error magnitude by depth and mean-subtracted path energy are distinct summaries; none alone certifies learning performance. Historical depth and noise cohorts, including their task-identity and conductance-validity records, are specified in Supplementary Section S2.

Contextual conflict and ancestry routing

In the branch-conflict task, one context-selected branch determines a logistic output while all B branches form ungated local eligibilities. Nonselected Fashion-MNIST views are replaced independently by opposite-class images with probability χ . Exact feedback is $\delta_i \mathbf{1}[b = b_i^*]$; shared feedback is δ_i/B , where δ_i is output error and b_i^* the selected branch. Twenty paired seeds cross $B = 2, 4, 8$ with eight conflict doses. The algebraically equivalent gated-point calculation and BP receive the same updates; cyclic derangement preserves rank and support while changing the recipient.

Under centered, symmetric class evidence, shared-mode compatibility is $s(\chi) = 1 - 2\chi(B - 1)/B$, with zero at $\chi_c = B/[2(B - 1)]$. This mean-field prediction is separate from a trained accuracy crossing. The latter is reported both by interpolation of the mean curve and by each seed’s first sampled at-or-below-chance dose, retaining non-crossings. Checkpoint re-execution reproduces endpoints and compares exact/shared fields at both common and own trained states (Supplementary Fig. S9C–E).

The ancestry factorial contains eight eight-dimensional blocks. Context c_i and binary label sign s_i are balanced, and inputs are $\mathbf{x}_{i,b} = a(b, c_i)s_i \mathbf{t}_b + \boldsymbol{\epsilon}_{i,b}$, with seed-specific unit teachers $\mathbf{t}_b$ and Gaussian noise $\boldsymbol{\epsilon}_{i,b} \sim \mathcal{N}(0, 1.1^2 I_8)$. The selected coefficient is one; sibling, same-half nonsibling and opposite-half coefficients are $-0.15, -0.45, -0.75$. A leaf permutation changes task-to-route alignment without changing the streams. For budget K , ancestry feedback has context-by-block entries $\Phi_{cb}^{(K)} = \sqrt{K/8} \mathbf{1}[g_K(\pi(c)) = g_K(b)]$, where g_K indexes contiguous groups and π maps streams to leaves. Thus every row has unit Euclidean norm.

The update is $\Delta \mathbf{w}_b = -(8\eta/N) \sum_i \delta_i \Phi_{c,b}^{(K)} \mathbf{x}_{i,\pi^{-1}(b)}$, with N training examples and rate η . Each seed supplies 2,048 training, 2,048 test and 1,024 switch examples. Full-batch descent uses rate 0.3, eighty training epochs and twenty switch epochs. Controls include deranged ancestry, depth-interleaved and random-sparse supports, dense rank- K fields and a training-informed PCA reference. The best matched control is the per-seed post-training maximum over the four non-PCA controls. Integer test counts preserve ties in the named Wilcoxon comparison; Holm correction covers four budgets. Alternative mean-sign-flip and historical BH analyses remain explicitly labeled in the Source Data indexed in Supplementary Table S12. The separate noisy-context encoder and cue-delay experiments are fully specified in Supplementary Section S3.

622 Update utility and prospective selection

623 Let the exact gradient be $\boldsymbol{\mu}$ and the delivered update estimator have mean $\mathbf{m}$ and covariance Σ_{route} at the current
 624 weights. For an L -smooth loss and a nonnegative step η , expected loss after the update is bounded above by $\mathcal{L} -$
 625 $\eta \boldsymbol{\mu}^\top \mathbf{m} + (L\eta^2/2)(\|\mathbf{m}\|^2 + \text{tr}\Sigma_{\text{route}})$. Optimizing this expression gives utility $U = [\boldsymbol{\mu}^\top \mathbf{m}]_+^2 / [2L(\|\mathbf{m}\|^2 + \text{tr}\Sigma_{\text{route}})]$,
 626 with zero for a zero denominator. Utility certifies one-step progress under the stated moments; it is not a long-horizon
 627 ordering.

628 Synthetic quadratic screens isolate signal retention, noise rejection and fixed gains over sixteen Haar coordinates,
 629 without a forward neuron (Supplementary Fig. S2). Their matched signal/noise hierarchy is a designed crossover.
 630 The trained factorial’s retrospective utility uses actual full-batch updates and a global logistic-curvature bound;
 631 choosing maximum bandwidth matches its selection success. The separate prospective linear experiment retains all
 632 12,800 fits, the initialization selector’s loss to fixed maximum-budget and rank-only baselines in every seed, and
 633 follow-ups forecasting evolving weights and noise. The 25,600-fit dynamics extension and observed-context-count
 634 control are reported in Supplementary Figs. S33–S34, with sealed choices, complete methods and computational
 635 costs.

636 Interaction compatibility, fixed profiles and learned spectra

637 For a subtree leaf set S , C_S arranges the target’s Fourier interaction coefficients by products inside and outside S ; C_S°
 638 removes the constant inside row. Scalar multi-affine trees represent the target exactly if and only if every centered
 639 cut has rank at most one. If $\mathcal{S}(T)$ contains all proper cuts of tree T and singular values are ordered from largest to
 640 smallest,

$$\text{MSE}(f, \hat{f}_T) \geq B(T; f) := \max_{S \in \mathcal{S}(T)} \sum_{j \geq 2} \sigma_j(C_S^\circ)^2. \quad (11)$$

641 Overlapping cut tails cannot be added as a lower bound. Supplementary Section S4 gives the constructive proof,
 642 minimum-depth algorithm and its relation to hierarchical tensor representations. The twelve-tree census, noisy-
 643 query estimator, end-to-end learning pipeline, conductance grouping controls and Boolean truth tables remain in
 644 Supplementary Figs. S11, S13–S14, S17 and S35. The theorem’s assumptions do not extend automatically to general
 645 conductance ancestors.

646 The matched credit experiment uses twenty fresh seed blocks after three development seeds select rates with
 647 equal tuning budgets. Matching and quartic conditions share tree, leaf assignment, initialization, inputs, additive
 648 noise and minibatch order. Exact, unit, initially calibrated and sign-only profiles use the same local eligibilities;
 649 calibration averages initial path derivatives over 256 unlabeled examples. Selected and common-rate outcomes are
 650 retained for SGD and Adam, giving 432 development and 1,200 fresh fits. Algebraic coefficients remain within
 651 $[-2, 2]$, and all bound contacts are recorded. The original fits stopped at 1,024 steps with seven saved checkpoints.
 652 A balanced extension retained all 720 unique algebraic trajectories at the selected/common-rate union, replayed
 653 every original checkpoint and continued to 16,384 steps. Its terminal and validation-selected results are descriptive
 654 analyses of the same seeds (Supplementary Section S4). Noisy held-out NMSE divides by clean target variance: one
 655 for matching and nested, one-half for quartic. All have noise variance 0.0225. The primary Adam contrast subtracts
 656 the pairwise calibrated-minus-exact gap from its quartic counterpart.

657 For evaluation-example-by-site path matrix Q , the best fixed profile is the leading right singular vector. Its capture
 658 is $\sigma_1(Q)^2 / \sum_j \sigma_j(Q)^2$, an oracle reconstruction with example-dependent amplitudes. Additional diagnostics evaluate
 659 uniform, calibrated and ancestry profiles, loss-residual weighting, eligibility-weighted updates and root-mean-square
 660 normalization of each site. Participation-ratio effective rank is $(\text{tr}C)^2 / \text{tr}(C^2)$ for uncentered $C = Q^\top Q / N$. At
 661 fixed weights and inputs, changing labels leaves the path field unchanged; eighty relabeling controls verify this.
 662 Independent checkpoint analysis and faithful replay of 600 earlier fits provide numerical and trajectory checks.
 663 Complete definitions, pairing audits, optimizer effects and the positive-conductance boundary are in Supplementary
 664 Sections S4–S5.

665 Conductance trees with context-dependent feature tuning

666 The seven-compartment model uses identity branch outputs in Eq. 9. Two independent $z_j \sim \mathcal{U}[-2, 2]$ generate
667 positive inputs $\exp(\pm z_j)$ for both subtrees. Every terminal has two excitatory and two inhibitory gains; binary
668 context inhibits one proximal compartment. All 24 conductances are positive and parameterized by logarithms.
669 Students start from terminal gains one, proximal inhibitory gains 15, daughter couplings 20 and 0.2 per subtree, and
670 soma couplings four. These proximal and coupling values match the teacher’s nominal scaffold, not its realized
671 weights. Independent Gaussian log perturbations have SD 0.4 in students and 0.15 in teachers. Full generators and
672 eligibility derivatives are given in Supplementary Section S5.

673 The local-gate follow-up used twenty new paired seeds and nine rules at Adam rates 0.01, 0.03 and 0.1, giving
674 1,080 trajectories. Each task supplied 2,048 training, 1,024 validation, 4,096 test and 512 unlabeled calibration
675 examples. The primary rate 0.03 was fixed from earlier work, not selected on these seeds. Minibatches contained 128
676 examples; logarithmic conductances stayed within $[-7, 7]$. Minimum validation NMSE selected states within fixed
677 4,096- and 16,384-update windows. The somatic error was $(V_s - y) / \text{Var}_{\text{train}}(y)$; NMSE uses clean target variance.
678 At a terminal with parent p , the hard gate multiplied this error by $\mathbf{1}[a_p = 0]$, where proximal inhibitory activity a_p
679 was zero or ten. Proximal and somatic multipliers remained one. The continuous gate used $1/(1 + g_p^I a_p)$. Both
680 combined these signals with the same exact local eligibilities and accessed no exact path derivatives.

681 Controls used exact credit, unit and initially calibrated broadcast, two weighted distal oracle profiles with unit
682 proximal credit, three oracle profiles including a proximal column, reversed distal gating, and gating at proximal
683 sites as well. Two primary paired sign-flip tests—opposed broadcast-minus-gate error and its opposed-minus-
684 aligned difference—were Holm-adjusted together. Other comparisons and 95% whole-seed bootstrap intervals were
685 descriptive. The protocol was committed after exposure to exploratory panel results but before execution on the new
686 seeds. Earlier development, wider-bound controls and new local-gate controls are in Supplementary Figs. S18–S20.
687 Full rule definitions, gradient-cancellation diagnostics, validation and outcomes are retained in Supplementary
688 Section S5 and its Source Data.

689 Serial depth, sensor alignment and optimization

690 Physical-depth networks contain 64 somata and eight nonsomatic modules each. D1–D4 reallocate these modules
691 across branching vectors $[8]$, $[2, 3]$, $[2, 1, 2]$ and $[1, 1, 2, 2]$ in the three- and four-tier comparisons; the two-tier D2 tree
692 uses $[4, 1]$. Contacts and resources are matched; serial composition changes. Grouped-point controls connect every
693 module directly to the soma. Raw-additive controls retain the graph without division, while flexible MLP controls
694 are approximately parameter matched.

695 Excitatory blocks carry class evidence multiplied by lognormal nuisance factors. Inhibitory blocks sense those
696 factors with alignment α : zero removes trial-specific information while preserving the sensor distribution, and one
697 reports the factor. The generator floors noisy input rates before multiplying them by gains. Nested, flat and local-ratio
698 families distinguish distributed hierarchical support from generic factor cancellation and locally available ratios. Full
699 generators, floors, sensor-placement controls and contact inventories are defined in Supplementary Section S6.

700 Ten paired seeds support each original cohort. The initial maximum is 180 epochs with patience thirty; validation
701 selects model states. BP and LocalCA optimizer recipes differ in learning rates and parameter-group treatment, so
702 their endpoint gaps are reported separately. Exact-path LocalCA computes the exact selected gradients. An audit of
703 one hundred original trajectories verifies every configuration and final metric against its source: all fifty D3 fits across
704 the five relevant credit/optimizer arms reached the epoch cap with validation loss still falling. The original depth
705 contrasts are therefore fixed-budget outcomes. Sixty source-consistent restarts raise the maximum to 600 epochs,
706 retaining all fifty D3 fits and ten exact-BP D1 references. Within each restart, minimum validation loss selects states
707 in both budget windows. All fifty D3 fits and two D1 references reach the larger cap; eight D1 references stop early.
708 D3 validation loss continues to decrease, and its selected states are near the cap. Complete histories, accuracy and
709 loss contrasts, saved-state checks and source-concordance discrepancies are retained in Supplementary Section S6.

710 Anatomical dictionaries and common-mode controls

711 Observed MICrONS skeletons and mapped contacts define soma-rooted branch-to-branch segments. Initial eight
712 and disjoint 47-cell cohorts come from the same mouse; Pinky is a second mouse, with ten of twelve selected cells
713 meeting the inherited typed-contact criterion. Each subtree route contains the input-bearing descendants of its origin.
714 Contact labels are direct where available and otherwise proxy-based as specified in Supplementary Section S7.
715 Compression preserves path length but only approximates resistance; direct-label, mapping and geometry sensitivities
716 are retained.

717 The modeled passive perturbation field consists of reciprocal passive-cable gradient changes induced by small
718 focal shunts. Every corrected dictionary receives one constant column and $K - 1$ spatial columns. Ancestry selection
719 uses inhibitory fraction times excitatory-area-weighted descendant fraction without field outcomes. Depth bins
720 are reparameterized with the same constant while preserving their span; random routes, independent site shuffles
721 and degree–depth surrogates provide controls. The weighted common component is removed before computing
722 residual-energy capture. A common-constrained SVD oracle supplies the rank-budget ceiling. Numerical rank,
723 coverage and nonzero density are reported separately; equal column count alone does not match those resources.

724 The original eight cells are post hoc development for common-mode correction. The protocol was frozen before
725 evaluating the new passive-response outcomes in v661 and Pinky. Two hundred randomized controls are averaged
726 within each cell. At eight columns, all 47 v661 cells and eight Pinky cells qualify; other eligible budgets are
727 retained without substituting for missing eight-column outcomes. Paired cell-bootstrap intervals and Holm-adjusted
728 comparisons describe within-cohort consistency, not animal-population replication. The older route-generated
729 positive controls use separate fields and endpoint definitions (Supplementary Section S7).

730 Focal shunting and the electrotonic boundary

731 Up to sixteen focal sites per cell span depth quartiles and require at least three excitatory descendants and three
732 comparison sites. Normalized E/I scales are 0.35 with reversals one and -0.2 . Shunt doses are 0.25, 0.5, one and
733 two times local membrane-plus-synaptic conductance. A matched injection supplies first-order current $\kappa_k(E_I - V_k)$
734 without changing the conductance matrix. A separately solved somatic current restores baseline voltage and is held
735 fixed during differentiation. The squared-loss target is one normalized unit below baseline; finite differences check
736 the adjoint gradients.

737 Localization subtracts depth-matched off-route from descendant medians of $|\log[(|\gamma'_i| + \varepsilon)/(|\gamma_i| + \varepsilon)]|$, with
738 $\varepsilon = 10^{-15} \max(1, \max_i |\gamma_i|)$. Because this is unsigned, signed gradient changes are reported separately. Adjoint and
739 driving-force substitutions in $\gamma_i = q_i(E_E - V_i)$ distinguish transport from local voltage effects. Sites and relation-
740 reassignment templates are nested within cells. The passive ancestry-partition identity and its dictionary qualification
741 are derived in Supplementary Section S8.

742 Physical calibration uses cylindrical axial conductance $\pi r^2/(R_a \ell)$ and leak $2\pi r \ell/R_m$, expressed in nS, with
743 spherical soma area $4\pi r^2$. Here r and ℓ are segment radius and length, R_a axial resistivity and R_m specific membrane
744 resistance. Sweeps include $R_a = 100, 150, 250 \Omega \text{cm}$ and $R_m = 300\text{--}30,000 \Omega \text{cm}^2$, fixed and locally normalized
745 doses, and background conductance. The disjoint calibration cohort contains 235 eligible sites in 45 cells. Geometric
746 fallbacks and dose definitions are specified in Supplementary Section S8.

747 The weak-channel check uses $R_a = 150$, $R_m = 1,000$ in those units and background conductance equal to leak.
748 Median maximal Na, K, Ca and HCN conductances are 0.06, 0.10, 0.025 and 0.05 times leak; NMDA is 0.15 times
749 excitatory conductance. Sixty-four draws per cell passed the predefined steady-state checks. Responses use rank-one
750 updates of the fixed baseline Jacobian, including voltage-gate derivatives, followed by somatic state matching. They
751 are local linearizations rather than newly solved nonlinear equilibria at each dose (Supplementary Fig. S29D,E).

752 Measured-response prediction and geometric diagnostics

753 The seven-target cohort includes thirteen eligible scans. Trial responses are the mean of the supplied ROI fluorescence
754 traces in `RoiResponseSeries` over each stimulus interval; our analysis did not deconvolve the traces or apply a
755 reliability correction. Repeated responses are averaged by stimulus condition for the ancestry-similarity analysis;

756 partial rank correlations control Euclidean separation and soma-to-contact path-length differences. Offline learning
757 holds out stimulus identities. Training-only percentile transforms produce nonnegative presynaptic proxies, and each
758 partner enters at its dominant mapped contact segment. A linear somatic readout predicts the standardized target
759 response. Only input-conductance credit differs across rules; readout and bias gradients remain exact.

760 Four-route dictionaries are selected without target responses and receive the projected baseline transfer profile
761 multiplied by the scalar readout error. Supplementary Section S9 gives the delivery and reconstruction equations;
762 Fig. S31 shows the realized supports and reconstruction diagnostics. Realized supports are reconstructed directly
763 from saved segment identifiers. The supplementary support matrix uses the scan with median mapped-input
764 count, with identifier-based tie breaking; all scan supports are available. At common exact-trained states, field and
765 eligibility-weighted update reconstruction compare the fixed profile, per-trial field projection and the best within-span
766 update. These are oracle diagnostics rather than newly trained rules. Reproducibility comparisons report endpoint
767 discrepancies across 130 exact-rule re-executions.

768 Ridge regularization is selected within training data for every outer split, using the same measured predictors.
769 Observed-input removal, predictor noise and reliability masks are separate sensitivity analyses with no held-out
770 information used for selection. Ten split/initialization replicates are averaged within scans and then within seven
771 targets. Imposed-alignment quadratic fields and the six-animal external signed-credit reanalysis are separate datasets
772 and tests, described in Supplementary Sections S7 and S9, respectively.

773 **Statistics and computational provenance**

774 Confirmatory choices precede outcome inspection but are not described as public preregistration. Pilots, post hoc
775 analyses and same-seed reruns are identified separately. Effect sizes accompany experiment-specific 95% intervals,
776 or adjusted 97.5% intervals where prespecified. Tests are two sided unless a directional protocol states otherwise.
777 Named multiplicity families and exact tie treatment are retained in Source Data; null results imply equivalence only
778 when a margin is tested.

779 For the sensitivity simulation, a positive-semidefinite shared-path covariance generated joint tuning across the
780 102 observed presynaptic roots. We mixed ancestry-correlated and independent reliable tuning, then added noise
781 calibrated to each recording's split-half reliability and actual repeat counts. Four thousand paired datasets at each
782 of 21 signal fractions were analyzed with the original partial-rank statistic and an exact seven-target signed-rank
783 test. Detection required two-sided $P \leq 0.05$ and a positive mean effect; Wilson intervals describe Monte Carlo
784 uncertainty. The model and analysis were frozen before simulated outcomes. Supplementary Section S9 gives the
785 covariance, reliability calibration, dependence assumptions and exact test resolution; the released pair-level inputs
786 permit reproduction.

787 Independent units are paired training seeds, reconstructed cells, postsynaptic targets or external animals. Sites,
788 contacts, scans, splits, checkpoints and randomized draws remain nested. Seed counts reflect the experiment's
789 purpose; no prospective parametric power calculation was performed. Source hashes certify retained inputs and
790 implementations, not unavailable historical runtime states. The matched credit and new conductance studies recorded
791 Python 3.10.13, NumPy 2.2.6 and pandas 2.3.3. The new MNIST controls used PyTorch 2.9.1/CUDA 12.8 and
792 torchvision 0.24.1 with NumPy 2.2.6; the depth extension used Python 3.10 and PyTorch 2.9.1. The isolated reviewer
793 environment was validated with NumPy 1.26.4, SciPy 1.15.3, PyTorch 2.9.1, torchvision 0.24.1 and Matplotlib
794 3.10.9. Cohort-specific runtime records distinguish executed experiments from this installation check. Supplementary
795 Section S10 consolidates statistical families and release records; the question-specific sections retain their exclusions
796 and numerical checks.

797 OpenAI Codex assisted with code, statistical and consistency audits, figure preparation, literature checks and
798 language editing. The authors remain responsible for design, interpretation, numerical accuracy, citations and the
799 final manuscript.

800 **Ethics and data reuse**

801 No new experiments involving animals or human participants were performed. All biological analyses reused public
802 MICrONS, DANDI, and published animal source data. The source studies describe the original data collection and
803 institutional approvals [22, 42, 60].

804 **Data availability**

805 MICrONS anatomy and connectivity are available through the public CAVE datastack minnie65_public. The
806 primary cohort used materialization 1822, skeleton-service reconstructions, and the synapses_pni_2 table. The
807 disjoint sensitivity cohort used the public v661 static release. Secondary target-structure labels used only in the
808 full-contact analysis came from the in-development synapse_target_predictions_ssa_v2 table. These labels
809 are distributed only as derived data and remain subject to the source-data access terms. All principal structural
810 findings were repeated without them.

811 The independent-animal replication used the MICrONS phase-1 layer-2/3 Pinky v185 fixed meshes, soma-
812 valence table, and dense synapse table archived at Zenodo DOI 10.5281/zenodo.3710459 [61]. File sizes, Zenodo
813 MD5 checksums, and local SHA-256 checksums are recorded in the supplied manifest. Volumetric imagery is
814 archived in BossDB under DOI 10.60533/BOSS-2021-T0SY, and co-registered functional data are available as
815 DANDI:000402. The supplied manifest records the exact asset and table identifiers, and the source publication
816 describes the full data release [42].

817 Public benchmark datasets are available from their original repositories. The inhibitory-census tables and SWCs
818 are available with the source record for Schneider-Mizell et al. [60]. The Francioni et al. workbook is available
819 through the Source Data link associated with the Nature article [22]; its checksum and the sheet names used here are
820 recorded in the manifest. Derived cell and exclusion manifests, numerical source data underlying every figure, and
821 non-restricted summaries are supplied with this manuscript. The measured-response release includes the analyzed
822 partner-pair tables, contact geometry, recording identifiers and repeated-stimulus response arrays needed for the
823 sensitivity analysis. Authentication tokens and externally hosted raw caches are not redistributed.

824 **Code availability**

825 The accompanying reviewer software archive contains the complete training implementation, exact-gradient and
826 perturbation analyses, versioned configurations, tests, figure scripts, and byte-identified snapshots of the critical
827 training components under the MIT License. A separate Source Data archive contains the numerical data underlying
828 every display item. These archives are available to editors and reviewers with the submission. The same version
829 will be released publicly under the MIT License in a versioned repository and permanent archival record upon
830 publication.

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Author contributions

H.S. developed the computational framework, performed the analyses, and wrote the initial manuscript. M.R. contributed to the theory of local credit assignment, its implementation, and its interpretation. B.L.S. supervised the work and contributed to the biological framing, study design, and manuscript. All authors reviewed and revised the manuscript.

⁹⁹⁸ **Competing interests**

⁹⁹⁹ The authors declare no competing interests.